

DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY



DIPLOMA IN INFORMATION TECHNOLOGY
(DIGITAL TECHNOLOGY)

NotiHouse Alarm System

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DECLARATION

We hereby declare that the project report entitled “NotiHouse Alarm System” is our original work under supervision and guidance of Pn Nur Yahzelina Binti Hanafi. We further declare that the work reported in this project has not been previously and concurrently submitted for award or any other diploma in Polytechnic or other institutions. Cited text and references are from the sources we have referred to and made appropriate acknowledgment of.

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"NotiHouse Alarm System"

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ABSTRACT

The need for home security alarm systems nowadays is a serious demand. As the number of crimes are increasing every day, there has to be something that will keep us safe. We are all aware of the high-end security system present in the market but they are not easily available to everyone. We therefore intend to provide a solution by constructing a cost-efficient electronic system that has the capability of sensing the motion of the intruders.

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1.0 CHAPTER 1

1.1 INTRODUCTION

A security system is a system designed to detect intrusion – unauthorized entry – into a building or other area. Security systems can be used in residential, commercial, industrial, and military properties for protection against burglary (theft) or property damage, as well as personal protection against intruders. Security alarms in residential areas show a correlation with decreased theft.

This system uses an inexpensive ultrasonic Sensor. An ultrasonic sensor allows you to sense motion, almost always used to detect whether a human has moved in or out of the sensors range. They are small, inexpensive, low-power, easy to use, and don't wear out.

1.2 PROBLEM STATEMENT

There are many types of home security systems that range from simple alarms system to fully-monitored systems that provide total security and safety coverage for home. Research and experts agree that a professionally studied, alarm security system is by far the most effective home security solution.

- Existing alarm system on the market uses different sensors and traditional detection method which cost more but less reliable and easily manipulated.
- Trespassing and theft cases have been increasing by day due to the economic down turn. Most of the people did not want to leave their homes because they are afraid of losing their belongings.
- The owner does not know and unable to monitor the house condition when they are not at home.

  Most alarm security system uses a magnetic contact to detect breaches or intrusion. This has become less reliable since most people or intruders can get access to a powerful magnet which can be used to disrupt the magnetic contact.

1.3 OBJECTIVES

Security System have the capability to alarm/notify the user via notification through internet if the security system is triggered by external factors such as theft. It also prevent theft and burglary in the house by certain measures such as ultrasonic sensor. This system able to complete the project which cost less than retail to produce a more affordable alarm system. Having a NotiHouse security system make worry less aboutthe criminal breaking in and taking the items.

To develop a prototype that uses an ultrasonic sensor

- Ultrasonic sensor can detect intruder with Arduino IDE as the development tool.

To configure the alarm system using BLYNK APP to monitor device status

- Using the BLYNK app not to just active or deactivate thre system. But also track timeline of break-in and device status.But also control settings and check device status.

To alert the owner and authority through the application notification

- The app also send the notification to the owner and authority in the event of a buglary or when device is offline.

1.4 SCOPE

NotiHouse system is an ideal solution for smaller home, apartment and condominium. Small business office, factories or home often neglect the security requirement. Even in isolated location of the government agencies mostly neglected the security requirement. For example schools nowadays equipped with multimedia equipment. These are often locked in the lab or in the locked stores which is unsecured. The lost of this equipment is far more expensive than the simple security system.

There are TWO(2) categories of project scope which is User Scope and System Scope.

1.4.1 USER SCOPE

User	scope
House Owner	● Unlimited access and authorization ● Add or remove features from device ● Configure every setting
Admin	● Configure apps and device setting using the BLYNK app. ● Can arm and disarm the system through BLYNK app.
Family members	● Can arm and disarm the system through BLYNK app.

Table 1.1 User scope

1.4.2 SYSTEM SCOPE

1. NotiHouse is a project that uses BLYNK as an android platform for user interaction.

2. The NotiHouse application can be used anywhere. The application required an internet connection for access.

In Scope	Out Scope
➢ **To notify the owner through the application and notify direct to police station.** ➢ **Able to detect intruder movement using an ultrasonic sensor** ➢ **Activate and deactivate the system with a unique authentication code** ➢ **Manually switch on /off when power and internet are unavailable**	➢ **Unusable without internet connection** ➢ **Useless in event of blackout or power shortage**

Table 1.2 System Scope

1.5 PROJECT SIGNIFICANCE

NotiHouse is an alarm system designed to detect intrusion. Our system uses an ultrasonic sensor to detect breaches into the house. By using this system, the user will be notified in the event of alarm setting off caused by intrusion on their phone with internet availability. By using BLYNK app, the system will notify the user if there were any intrusion. This system is quite affordable compared to most security alarm system on the market. The cost for this project is planned to be Rm68 while the other security alarm system cost Rm288 and above excluding installation fees. Our project cost less since the IOT components such as Arduino ESP8266 and ultrasonic sensor has become more affordable over the years.

2.0 LITERATURE REVIEW

2.1 INTRODUCTION

In this project, there are three researches that have been made before proceeding to implement the project. When you look at your family and house, the first thing you want is their protection. And then the concept of an integrated system of home protection comes into the frame. If we look at various home automation systems over the time; they have always tried to provide home residents with reliable, easy, and secure ways of accessing their homes.

To build a cost-effective and efficient security automation framework, we can make use of IoT technologies. The systems available are either costly or proprietary so that a user could not afford them or they are forced to be loyal to a particular provider that inhibits the number of choices available to them and prevents them from investing in the security infrastructure required. Intruder detection systems based on IoT can have the security ability integrated with communication features to alert the intruder's owner even before the intrusion and avoid property damage. We will review the various methodologies and technologies that are used to solve the problem of intruders in the following paper.

2.2 e-KTP as the basis of home security system using Arduino UNO:

During the construction of the intruder detection system, there were several systems we encountered. The Arduino UnoR3, ESP8266 NodeMcu WiFi module, alarm, Red Light Emitting Diode (LED) and PIR Sensor are used in the' Arduino-based Wireless Motion Detecting System To build this device framework, Android Studio 3.1 software was used. The programming language used in the device application was Java.

The transmission of data to the consumer is parallel with the used alarm device. If the alarm device senses movements, applications which have been imported to users' smartphones may cause an alarm. A server-based approach is used by the system so that the user is bound to the Internet. The Arduino Uno microcontroller is used as a doormat in a door-based digital home protection system in the 'e-KTP as the basis of the home security system using Arduino Uno Radio Frequency Identification (RFID) system from e-KTP.

The data stored on the e-KTP card will be transmitted to the RFID Reader when the power is switched on and an e KTP card is present. If the check is ineffective, a "threshold" warning will be generated, the red light will be switched on, and an alarm will be generated to identify that the robbery has occurred in the building. To build a database for e-KTP on the internet, a XAMPP server was used.

2.3 IoT based Intruder Detection System Using GSM :

The Internet of Things (IoT) is a network of interconnected objects that can interact and share data. The combination of software and hardware developments has allowed the development of systems that can run remotely without the need for human interaction. The proposed device would not act like any of the currently available protective devices, which do not warn the consumer when an intrusion occurs. This device notifies the user when a disruption happens, so it does not produce any false alarms.

The hardware module and a mobile application make up the framework. When the hardware module senses any movement, it sends a warning to the app's users. As soon as the device is activated, it will send a message to the application, alerting it. The aim of this project is to build an intruder detection application using an AVR microcontroller device, conduct User Acceptance Testing and implement an external beta tester. Since our device is modular, the user can add as many hardware modules as they want.

The software can be used by either home or shop owners to receive alerts in the event of any unauthorized intrusions. In order to detect an intrusion, the sensors MC-38 (magnetic switch) and SW-420 (vibration sensor) are used to implement the device. The code created in the Arduino IDE was uploaded to the atmega 8A using an FTDI programmer. When the sensors are activated, a sim 800L warning is sent to the application. Android Studio was used to develop the software. The hardware device detects the intrusion and notifies the program successfully.

The device's versatility allows it to be incorporated into existing security infrastructure. The device's simplicity makes it perfect for scaling up to meet the needs of the consumer. The device's hardware is built in a modular manner, enabling it to be easily enhanced by connecting it to a wireless sensor network. The system can be upgraded by adding an internet module in addition to the GSM module. This would allow the computer to switch between the two communication interfaces easily when one is inaccessible, improving the system's overall performance.

2.4 Home security system using internet of things :

The Internet of Things (IoT) is a network of interconnected physical objects that can interact and exchange data without needing human interference. Since then the Internet of Things has expanded exponentially, with more than 12 billion connected devices now in operation with analysts expecting that number to hit 50 billion by the end of 2020. When it comes to home automation, security is a significant aspect. Home security is one of the most important aspects of home automation; if not the most important. Home security has advanced significantly in recent decades and will continue to do so in the coming years.

Home security systems used to mean getting an alarm that would go off if anyone broke in, but a smart safe home can do so much more. As a result, the main purpose of our work is to build a device that can send a warning to the owner and others in the event of an intruder break-in. The alarm would also be able to be stopped or started remotely using the owner's mobile. Users will be able to protect their homes by installing the device on their doors or windows and tracking activity through their smartphones. All of these internet-connected devices are part of the IoT infrastructure, which enables them to send and receive data from one another.

This is why it is advantageous to design the proposed security framework using an existing infrastructure. When a user is not present in the home to take action an alarm that sounds like the buzzer is useless. When the owner is gone, keep in touch with one another. They want to know that their home is safe from intruders and criminals when they are abroad, so they set up an IoT network with embedded electronics, sensors and apps at their home. As a result, the proposed system keeps the owner up to date on the security status of their home in real time. The built system warns the user when there has been a break-in, enabling the user to take appropriate action.

NO	Paper Title	Technologies used	Strength	Weaknesses
1	e-KTP as the basis ofhome security system using Arduino UNO	RFID	i)Only unique ID number can give access to open door ii) Web based loggingsystem	i)Metal and liquidscan impact signal
2	IoT based Intruder Detection System Using GSM	GSM	i) Alert warning toowner ii) Simple and easy implementation	i) Less efficient ii) Not reliable enough
3	Home security systemusing internet of things	Arduino UNO	i) Open source (Free) ii) Easy implementati on	i) Unusable withoutinternet ii) Useless in event of blackout or power shortage.

Table 2.1 Literature Review Comparison

2.5 CONCLUSION

Our project mainly concentrates on the application aspects of the numerous intruder system and also points out the various involvement of technologies used to make the intruder detection system practical for real-life application. We summarized the various systems that utilizes low-cost components while having potential for industry standard usage. We encourage researchers to think about home automation as one of the most critical aspects of home protection and to develop advanced sensing technologies to detect and secure homes from professional intruders.

As we progressed IOT architecture and sensors became more budget friendly. We can use different types of sensors such as vibration sensing accelerometer, reed switches, PIR sensors. Hence, we recommend IOT can bring the new heights in theft detection systems.

3.0 METHODOLOGY

3.1 INTRODUCTION TO AGILE DEVELOPMENT MODEL

The Agile model was primarily designed to help a project to adapt to change requests quickly. So, the main aim of the Agile model is to facilitate quick project completion. To accomplish this task agility is required. Agility is achieved by fitting the process to the project, removing activities that may not be essential for a specific project. Also, anything that is wastage of time and effort is avoided. In the Agile model, the requirements are decomposed into many small parts that can be incrementally developed. Agile model is the combination of iterative and incremental process models.

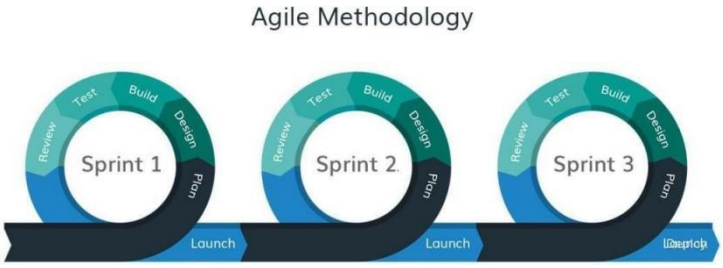


Figure 3.1 Agile Development Model

The reason for choosing this methodology is because of the nature of the system that is going to be developed. Agile model relies on working software deployment rather than comprehensive documentation. It is recommended that the development team size should be kept small (5 to 9 people) to help the team members meaningfully engage in face-to-face communication and have collaborative work environment. Agile development process usually deploy Pair Programming. In Pair programming, two programmers work together at one work-station. One does coding while the other reviews the code as it is typed in. The two programmers switch their roles every hour or so. It emphasizes on having efficient team members and enhancing communications among them is given more importance. It is realized that enhanced communication among the development team members can be achieved through face-to-face communication rather than through the exchange of formal documents. The Agile model adopts Iterative development. Each incremental part is developed over an iteration. Each iteration is intended to be small and easily manageable and that can be completed within a couple of weeks only. At a time one iteration is planned, developed and deployed. It reduces total development time of the whole project.

3.2 PHASE 1 : REQUIREMENTS

For the early stages, analysis will be made and one of them is requirement analysis. We have observed the existing systems that are being used to understand their features as well as the existing weaknesses and strengths present in the system. We have used these existing systems as a base to plan and design our system which is much more improved and secure. The requirement of the proposed system was gathered. The requirements included the functional and non-functional requirement which is essential for the development of the system.

Functional Requirements	**Non-functional Requirements**
- User are able to access the device from an application. - Owner could know if device is active or disarmed. - Connected via the public internet. - All code is signed and only trusted code will be executed. - All communications are encrypted and authenticated. - All access to resources must be authenticated and authorized.	- Make a small and compact device. - Can't be too resource intensive - Easily manageable. - All IoT applications run in a secure container

Table 3.1 functional and non-functional of system

Other than that, literature reviews for similar projects such as e-KTP Arduino Intruder Detection System, GSM Intruder Detection System, and Security System with Telegram Alert will be made. This literature reviews study in terms of user interfaces, features, suitable programming language, and methodology used. Reviewing these literatures would help us create a system that is suitable for the proposed system.

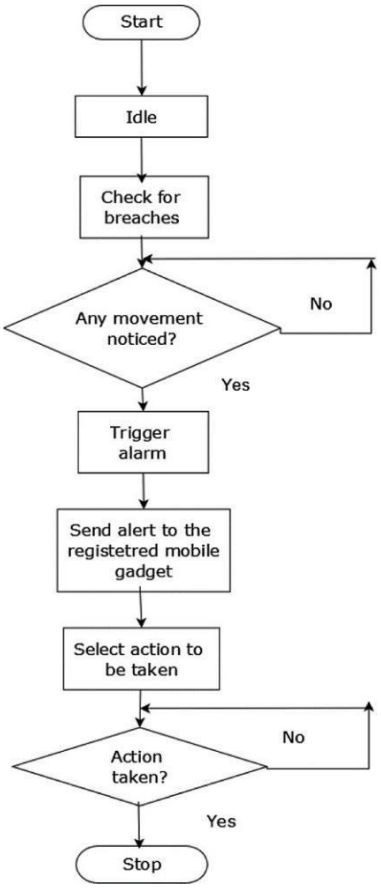


Figure 3.2 Flowchart of Proposed System

User-friendly interface, minimal usage of words and most importantly the arrangement of the application must be exact. The feature of the system which will be studied are being considered and would be included in the proposed system if the feature is usable and helpful. A system requirement document that consists of the requirement of the user will be created once the information gathered are complete.

3.3 PHASE 2 : DESIGN

3.3.1 LOGICAL DESIGN

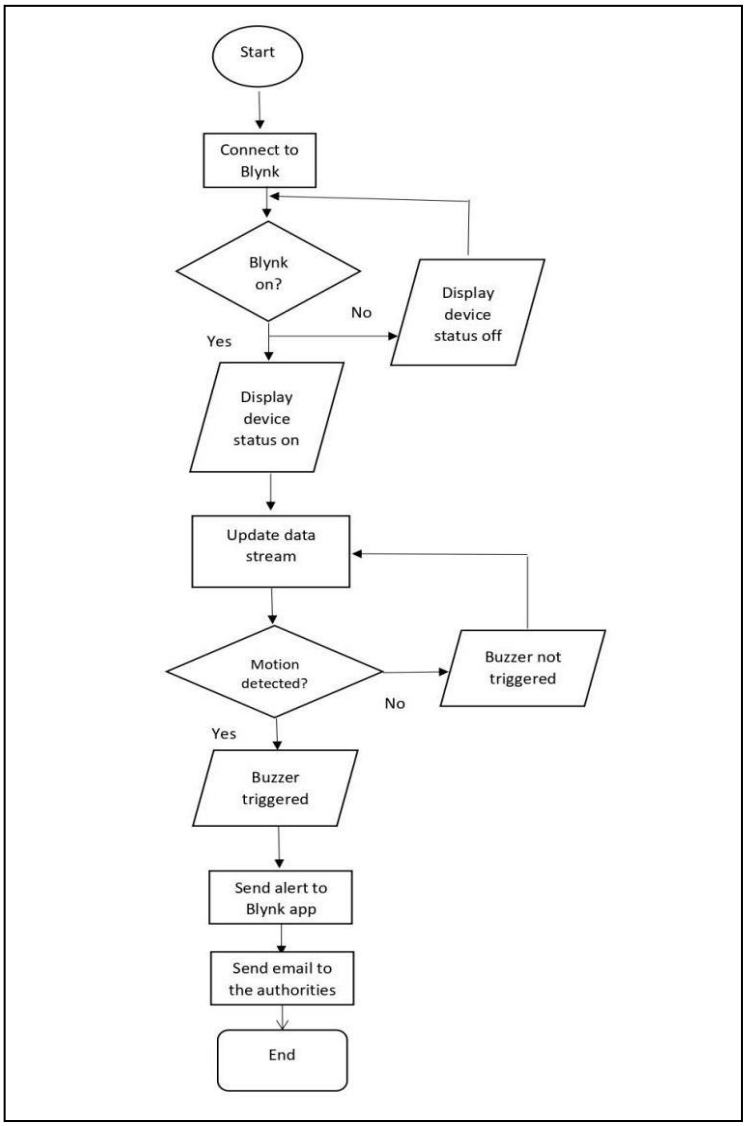


Figure 3.3 Flowchart of NotiHouse alarm system

Figure above show the flowchart for the NotiHouse alarm system. Firstly, the user must login in order to access the Blynk GUI. Then,the user must turn the Blynk on , to activate Blynk system. After that, the device will display the system is on. If there are any motion detected by ultrasonic sensor. Then, the buzzer will be triggered and the device will send alert to the Blynk app. The user will receive the push notification and alert email will be sent to the authorities.

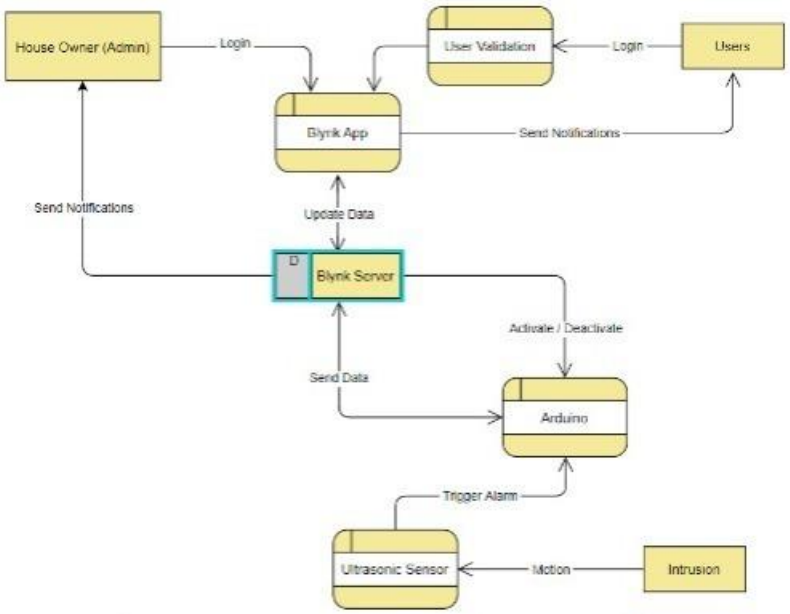


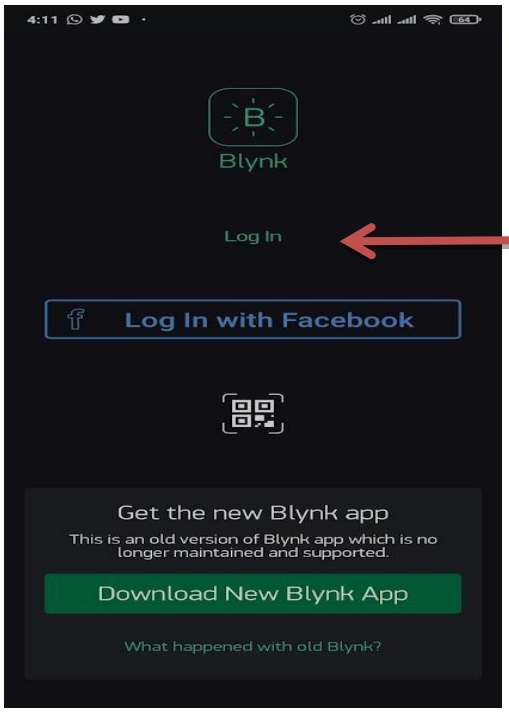
Figure 3.1.1 : Data flow diagram of NotiHouse alarm system

The figure above shows the DFD diagram for NotiHouse Alarm system. It is a graphical representation of a system which only use one process to represent the entire system. The purpose of this diagram is to display the expecting relationship between external entities who interact with the system such as the house owner(admin) and the user.

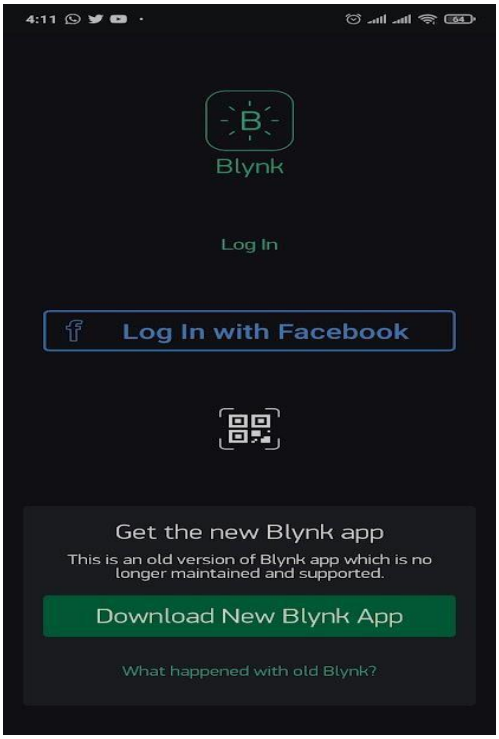
The house owner have full access and configuration to the system. While the user can only access the system but cannot configure the system and widget settings .The house owner can validate the user by sharing the new QR code and the owner also can revoke the user access by refresh the QR code. When an intrusion is detected, the house owner will receive the push notification from the Blynk app and also send email alert from Blynk server. While ,the user only will receive push notification from Blynk app.

3.3.2 PHYSICAL DESIGN

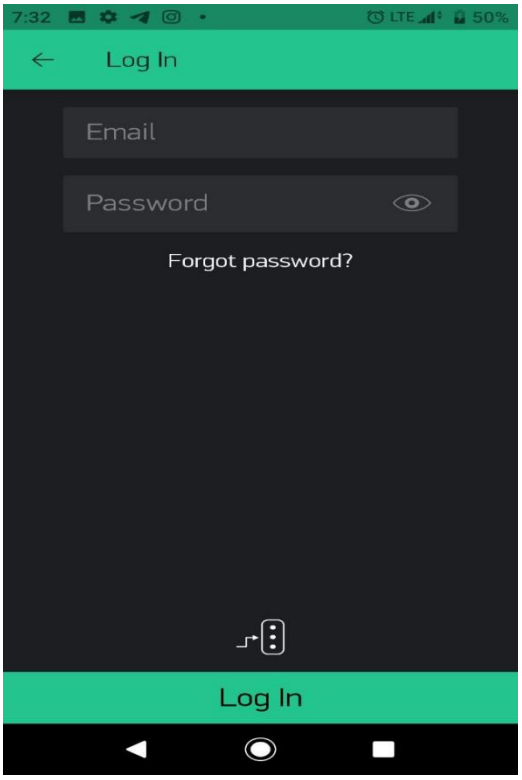
1. Launch Blynk application



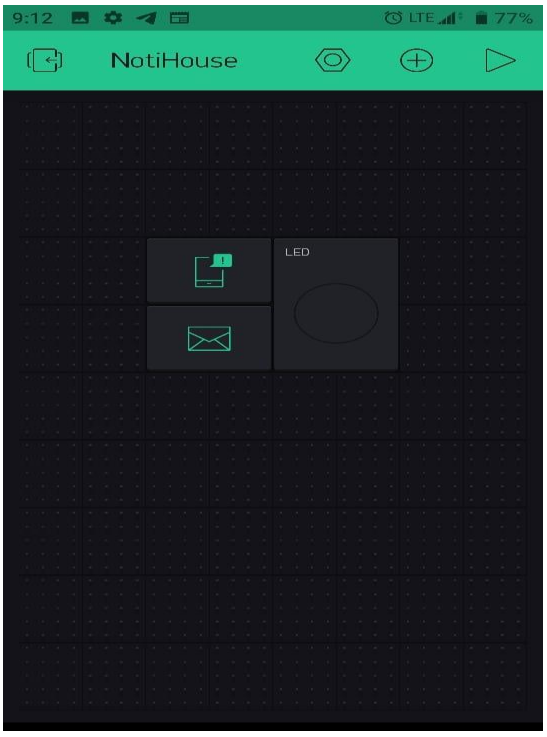
2. Click Log In button



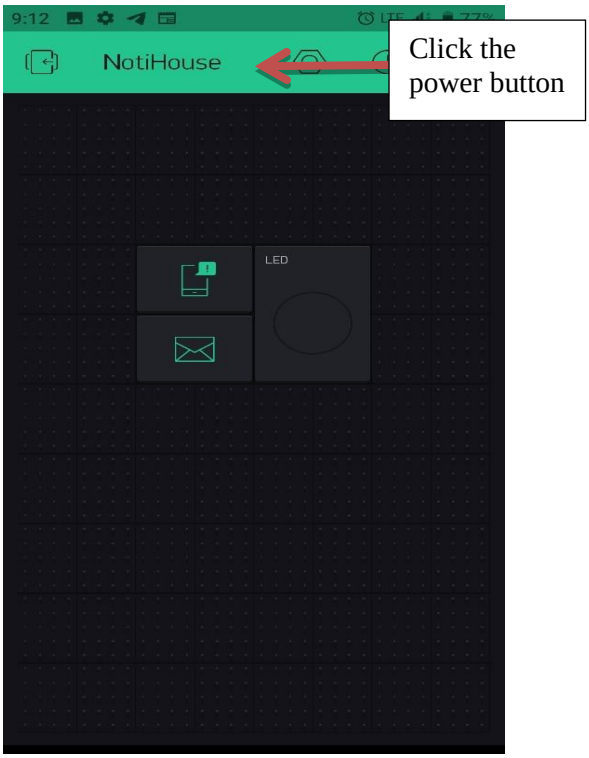
3. After clicking Log in button,then enter your username and password.



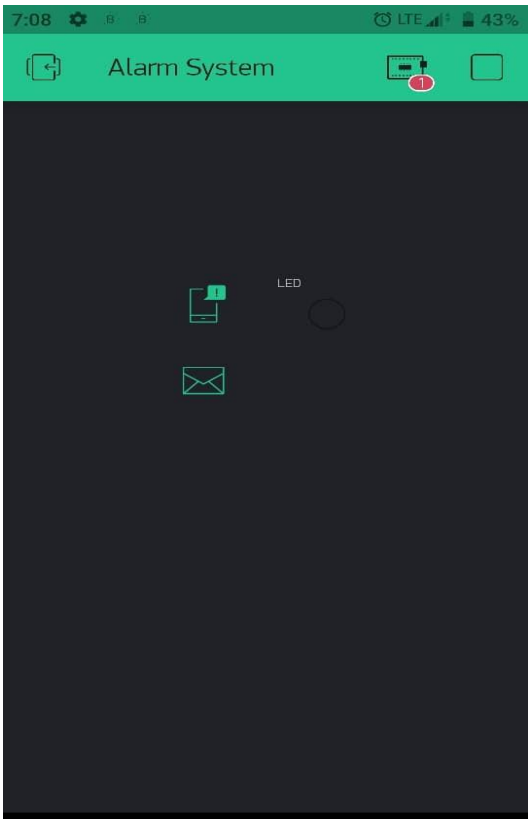
4. Blynk NotiHouse ui will appeared on the device



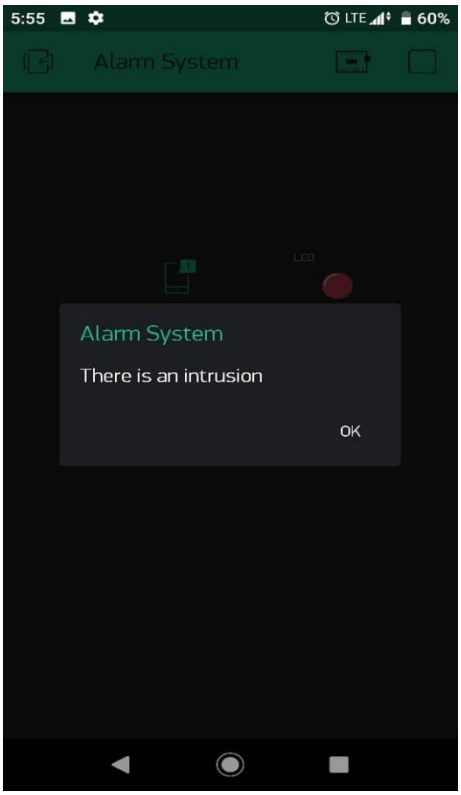
5. Click this button to activate the Blynk system.



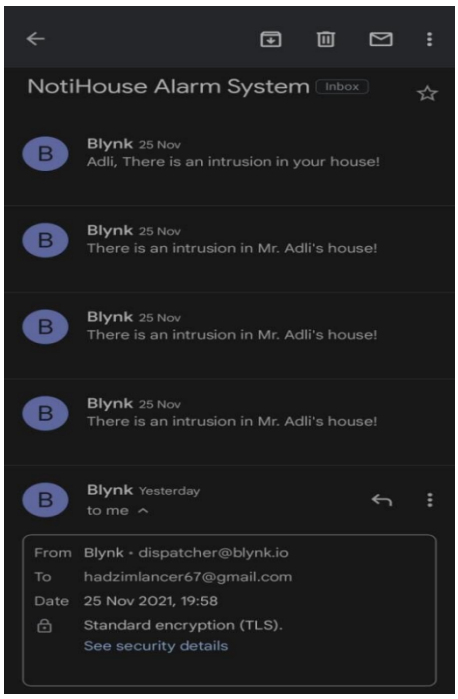
6. The Blynk will be activated



7. If the ultrasonic sensor detected the motion,then push notification will appeared on the user device.



8. The alert email will be sent to the authorities



3.4 PHASE 3 : DEVELOPMENT

Before we begin connecting the hardware, we have to get the ESP8266 set up by flashing the latest version of the firmware available for the module. This is because the chip comes with an older version of the AT command firmware pre-installed out of the box which cannot communicate with the Blynk libraries efficiently and will give an error with our code. To flash the latest firmware, download the ESP8266 flasher tool and the latest firmware from the internet which would be in the bin format and set up the ESP8266 to the Arduino Uno.

Once the ESP8266 has been flashed with the latest firmware, other components can be added to the configuration. For this we will need a breadboard to connect the microcontroller, ultrasonic sensor, buzzer and the ESP8266 using the jumper wires. The breadboard is used to interface between the various components available. It also makes it easy to connect multiple inputs to a single pin on the Arduino board.

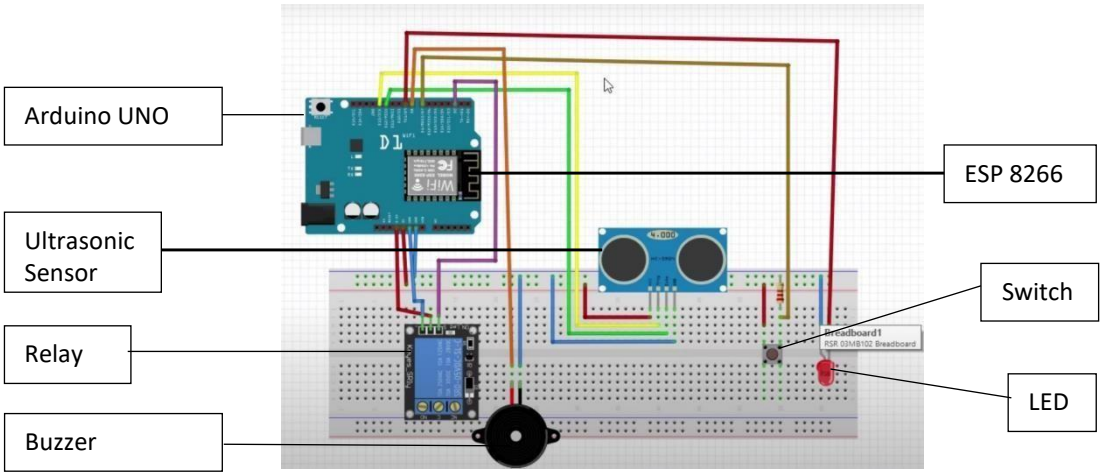


Figure 3.3 Sketch Diagram

Following sketch shown in Figure 3.3, which has been constructed using the Fritzing software shows how the components are supposed to be connected together using the breadboard and the jumper wires. The final configuration need not be identical to the given sketch, although the pins on each device needs to be connected to the same corresponding pins on the Arduino Uno board. The architecture diagram is shown in Figure 3.4.

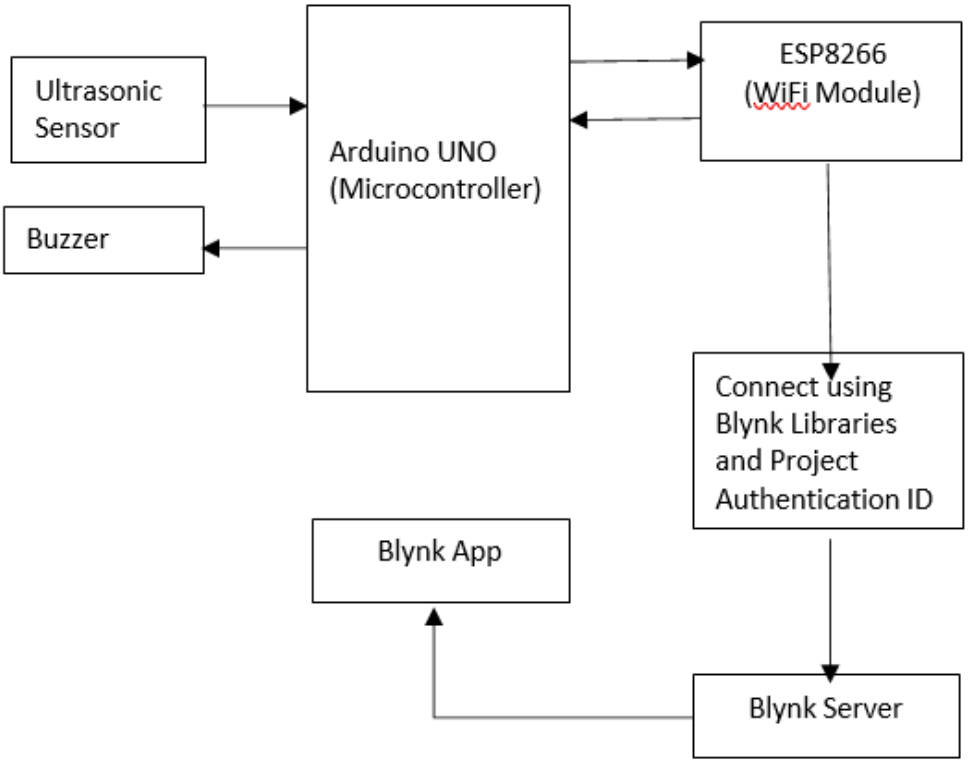


Figure 3.4 Proposed Architecture Diagram

3.5 PHASE 4 : CONFIGURATION AND CODING

After the user installs the Blynk app on the smartphone, an account has to be created in the app to access its services. The first time the app is opened, it will ask to either sign in or create an account. Create an account and add a new project to get started as given in Figure 3.5. Each project has its own authentication code which is used by the code to communicate with that particular model as provided in Figure 3.6. To interface with our components, we need to add widgets to our model. To add widgets press „+" to add to the model. The app provides a neat interface to add all the required widgets and setting them up according to the code as shown in Figure 3.7. The Blynk needs to be running in the background for the user to get real time notifications.

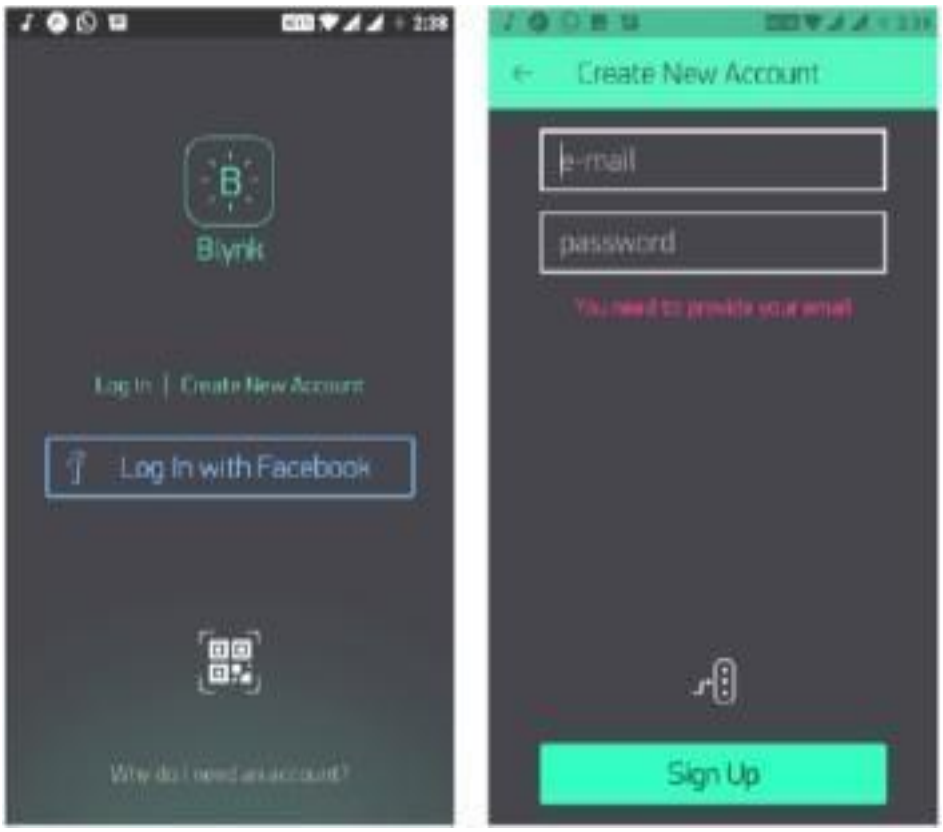


Figure 3.5 Setting up an account

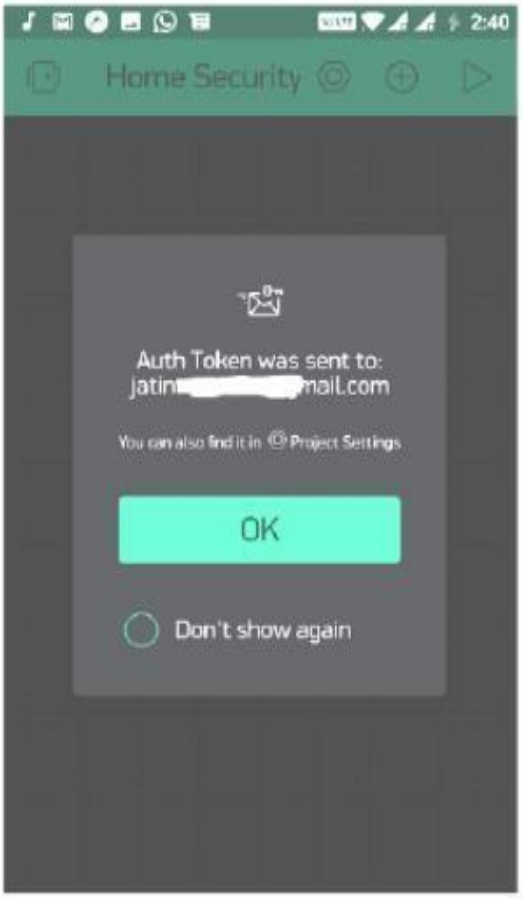


Figure 3.6 Authentication Code (Token)

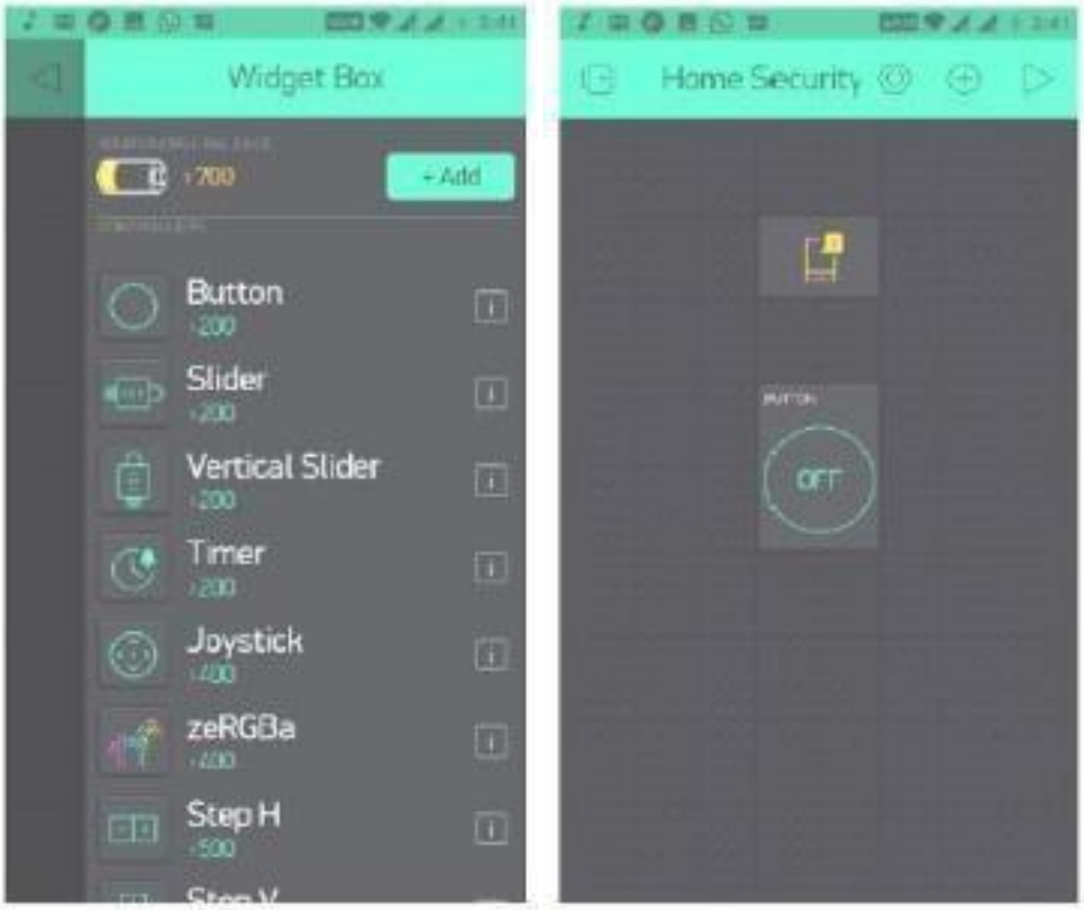


Figure 3.7 Adding Widgets

The development stage is where the database will be created based on the database design produced in the architecture diagram. Arduino code is written in C++ with an addition of special methods and functions. The Arduino Integrated Development Environment (IDE) is the main text editing program used for Arduino programming. It is where we'll be typing up our code before uploading it to the board you want to program. Arduino code is referred to as sketches. When you create a 'sketch' (the name given to Arduino code files), it is processed and compiled to machine language. Essentially, the IDE translates and compiles your sketches into code that Arduino can understand.

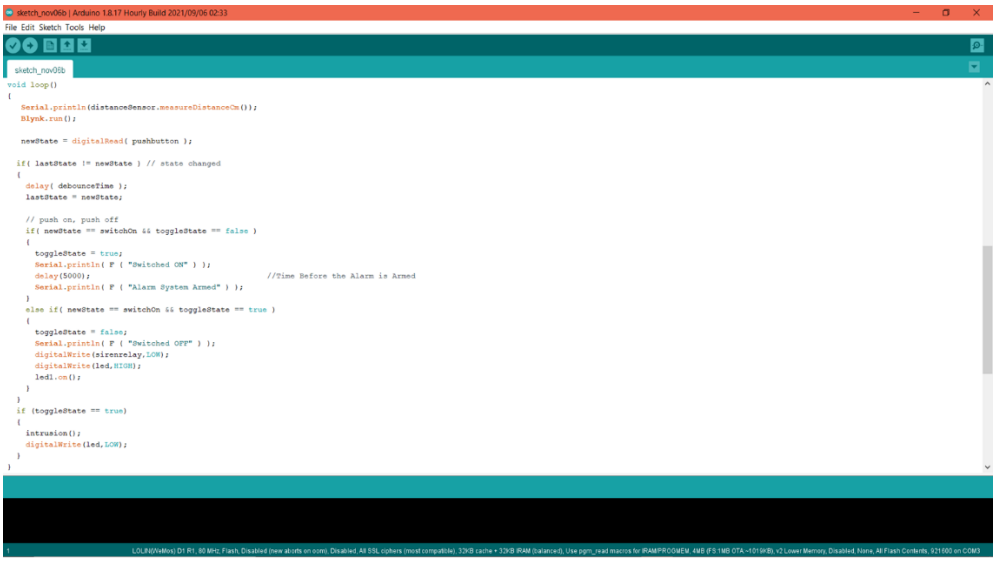


Figure 3.8 C++ written in Arduino IDE

Once your Arduino code is compiled it's then uploaded to the board's memory. Hardware can connect to Blynk Cloud (open-source server) over the Internet using hardware connectivity available on your board (like ESP32), or with the use of various shields (Ethernet, WiFi, GSM, LTE, etc). Blynk Cloud is available for every user of Blynk for free. Blynk Server is an Open-Source Netty based Java server, responsible for forwarding messages between Blynk mobile application and various microcontroller boards and SBCs (i.e. Arduino, Raspberry Pi. etc).

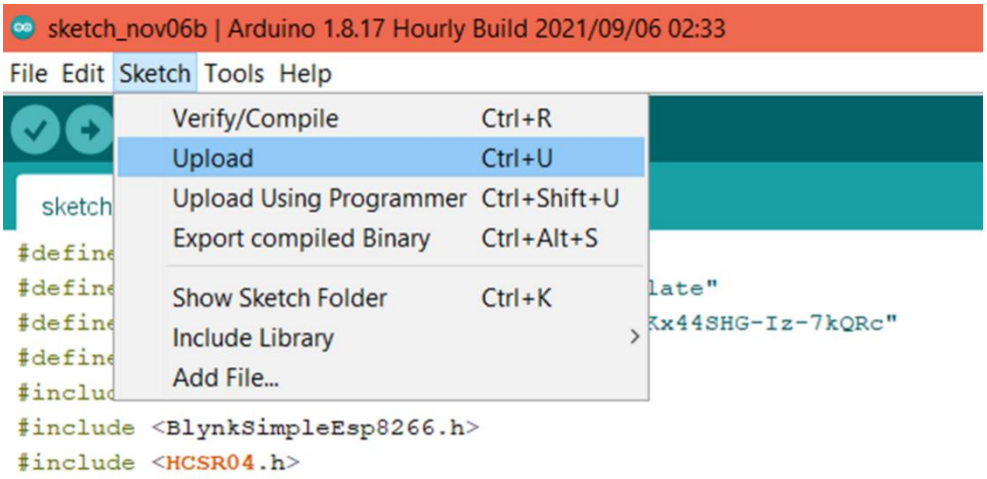


Figure 3.9 Upload sketch code into board

3.6 PHASE 5 : INTEGRATION AND TESTING

Testing will be performed as planned once the system has been developed. The test plan will be designed for when the system has been developed. The test plan must contain the procedures that are required to perform testing.

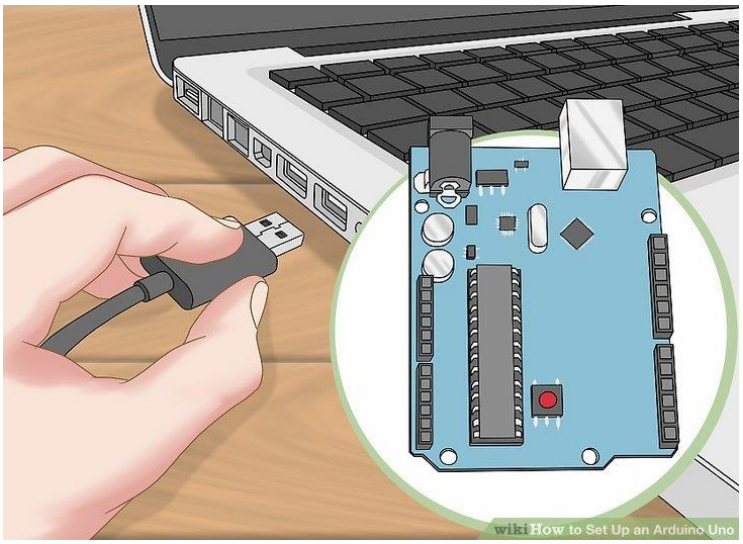


Figure 3.9 Plugin Arduino board to PC

The developer will be required to test the system as a different member will register input to the device. The developer must input the test data written in the test plan and check whether the output is as expected. If the output is not as expected, it will be recorded so that the evaluation of the system can be done.

3.7 PHASE 6 : IMPLEMENTATION AND DEPLOYMENT

In this phase, the system has met the requirement of the developer, and the devices with all its functions operating as intended which helps the end user to be more informed on the safety and security of their home. Therefore, the system is ready for deployment and ready to be installed by the user. The proposed location for server deployment is in the Blynk Cloud which is an (open-source server) over the internet. This to ensure the server uptime and reliability.

This setup will save the cost of maintenance and installation. The whole process of development must be documented so the system could be easily maintained in the future. Simultaneously, it would also help the user to understand how the system operates.

3.8 PHASE 7 : EVALUATION

In this final phase, evaluation of the system begins. Evaluation carried out will check on the system functions. From the testing phase, we can find out if the system meets the user requirement. We can then decide whether the system can to go into the next iteration or to perform incremental on the existing functions.

Other things to consider now is if the user wants more functions. If so, then the system will go into another iteration where the development will go back to planning phase. Enhancement the existing function will also lead to the development to go back into planning phase.

3.9 COMPONENTS INVOLVED

3.9.1 HARDWARE REQUIREMENT

Minimum hardware specification to develop NotiHouse System are as follows :

Parts	Minimum Requirement
Processor	3.0 GHz or higher
Graphics Card	Intel Iris Xe or higher
Hard Drive	20GB of available disk space
Memory	4GB or higher
Display	1280 x 720 panel or higher

Table 3.8.1.1 Hardware specifications to develop the system.

3.9.2 SOFTWARE REQUIREMENT

Minimum software specification to develop NotiHouse System are as follows :

Parts	Minimum Requirement
Operating System	Microsoft Windows 8 or later
Web Browser	Google Chrome, Microsoft Edge or Opera Browser
Server	Blynk Cloud Server
IDE	Arduino IDE
Runtime	Microsoft Visual C++

Table 3.8.1.1 Software specifications to develop the system.

Minimum software specification to run NotiHouse System are as follows :

Parts	Minimum Requirement
Operating System	Android 8 or later
Web Browser	Google Chrome or any Browser
Server	Blynk Cloud Server
Application	Blynk App
Account	Google Account / Gmail

Table 3.8.1.1 Software specifications to develop the system.

4.0 DISCUSSION

4.1 ADVANTAGES OF OUR PROJECT

NotiHouse project has many advantages that users can enjoy. NotiHouse is an alarm system that uses ultrasonic sensor and esp8266.This alarm system can help to monitor user house by detect any motion and send an alert if they are any motion detected.

Simple system

NotiHouse alarm system is a simple system. This is because users do not have to use card access which have a risks. For an example RFID alarm system, if the users forget to bring card access then they will not be able to access the system. The user also can set up this alarm easily. The user only need to charged the NotiHouse device then it will activated.

Reliability

NotiHouse alarm system is also a high-quality product. This is because NotiHouse alarm system using ultrasonic sensor as its main sensor. This sensor cannot be manipulated unlike other sensors such as magnetic contact sensors that can be manipulate by using another magnet.

Friendly user interface

NotiHouse GUI interface has a user interface that is user friendly and easy to use. The owner has full access to the mobile application's functions while the authorized user only can access the system but cannot configure the setting or widget functionality. GUI elements such as notification widget, email widget and led indicator are used to alert the user or owner about the intrusion detected.

4.2 LIMITATIONS OF OUR PROJECT

1. **Power supply**

NotHouse alarm system are fully electronic. So, it needed a constant power to keep it activate. Whenever a power shortage happens such as black out, NotiHouse device will not function. This is because this device did not have a spare power supply which is battery.

2. **Blynk app**

Blynk app is an IOT app that can connect with an IOT devices. This app is really useful and have many widget that can help the developer to create their own GUI by drag and drop. However, the use of this application is too expensive. This causes the use of widgets need to be reduced due to financial factors.

3. **Connectivity**

NotiHouse device uses an esp8266 as a main board. So ,by using this component ,we can connect with Blynk by using WIFI. When the WIFI is unavailable, the NotiHouse device will not connected with Blynk app. As a result ,the user and the authorities will not receive any alert notification from the device .

5.0 CONCLUSION AND RECOMMENDATIONS

5.1 SUGGESTION

NotiHouse is an alarm system that uses adruino uno and Ultrasonic sensor as a device component .NotiHouse alarm system have many advantages but it also have it's own disadvantages.Our suggestion are to add internal battery on the alarm system.This is because when blackout occur then the device still have a spare power supply.As a result,the alarm system will still functioning even when the device is not charging.Next suggestion is to use a cheaper iot application .This is because many iot applications need to be purchased to be use.The user also cannot express their creativity because of the limitation on the iot app due to the financial problem.

5.2 CONCLUSION

In a nutshell,our project has many functions that can help our client in the future.NotiHouse alarm system also meet the needs of our client.Our device are inexpensive,durable and has many function.We also provide app to the user.By using this app ,the user can monitor their house even when they are not at home.The most important thing is implementation of this project in present daywill effectively provide a peace of mind to our user.

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APPENDICES
GANTT CHART & COST PLANNING

GANTT CHART

Project Activities	Duration day	Start date	
Requirements	16	9/27/2021	
Discuss project title	5	9/27/2021	
Prepare introduction, problem statement, scope and objectives	2	10/2/2021	
Define methodology and cost planning	4	10/4/2021	
Research literature review	5	10/8/2021	
Design and development	21	10/16/2021	
Sketch project prototype	3	10/16/2021	
Design gui on BLYNK	2	10/19/2021	
Develop project prototype	10	10/21/2021	
Develop source code for prototype system	6	10/31/2021	
Configuration and coding	10	11/6/2021	
Configure app functionality	5	11/6/2021	
Implement code into prototype	5	11/11/2021	
Integration and testing	7	11/16/2021	
Unit testing	4	11/16/2021	
Integration testing	3	11/20/2021	
Implementation and deployment	10	11/23/2021	
Deploy the system	5	11/23/2021	
Install the project	5	11/28/2021	
Evaluation	11	12/3/2021	
Evaluate integration and testing result	11	12/3/2021	

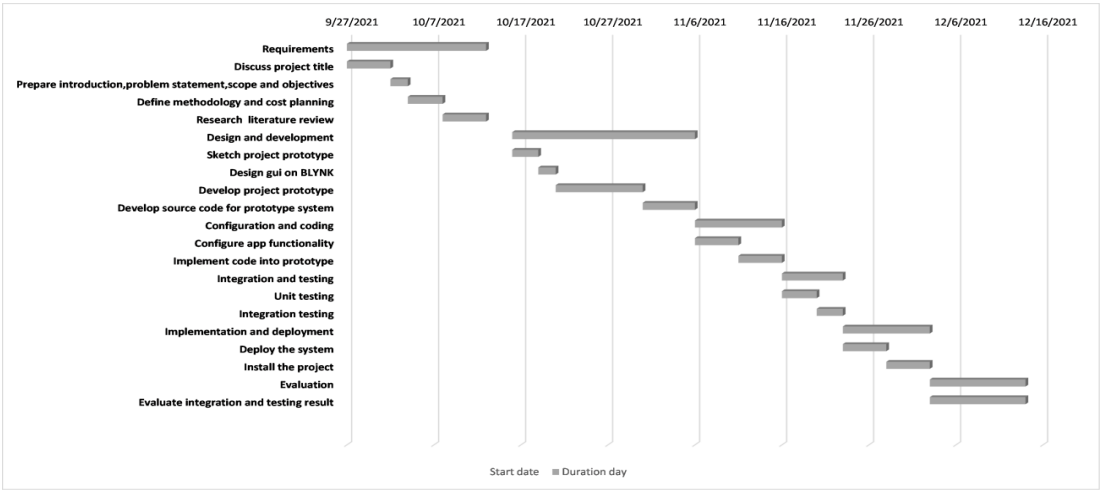


Figure : Project Gantt Chart

COST PLANNING

The Arduino security alarm system is an IOT project that uses many components. The components that we need are not free so we have made an estimation of the cost for our project. Our budget for this system electronic component is about RM55 and the estimation for shipping cost is RM18. Meanwhile, the cost for the software that we use is free such as BLYNK and Arduino IDE. The purpose of our cost planning are to save costs and prevent money from being wasted unnecessarily. The componentpurchased are as follows :

Component/item	Price(RM)	Quantity	Total(RM)
Arduino ESP8266	21	1	21
Relay	8	1	8
Buzzer	3	1	3
Button	1	1	1
Ultrasonic sensor	4	1	4
Led	2	2	4
Jumper male to male	5	1	5
Jumper female to male	5	1	5
Breadboard	4	1	4
			Rm55

Table: Cost estimation of hardware (Not included shipping cost)

Software	Specification	Cost
BLYNK	Version 1.3.3 Play store	Free
Arduino IDE	Version 1.8.16 windows 64-bit	Free

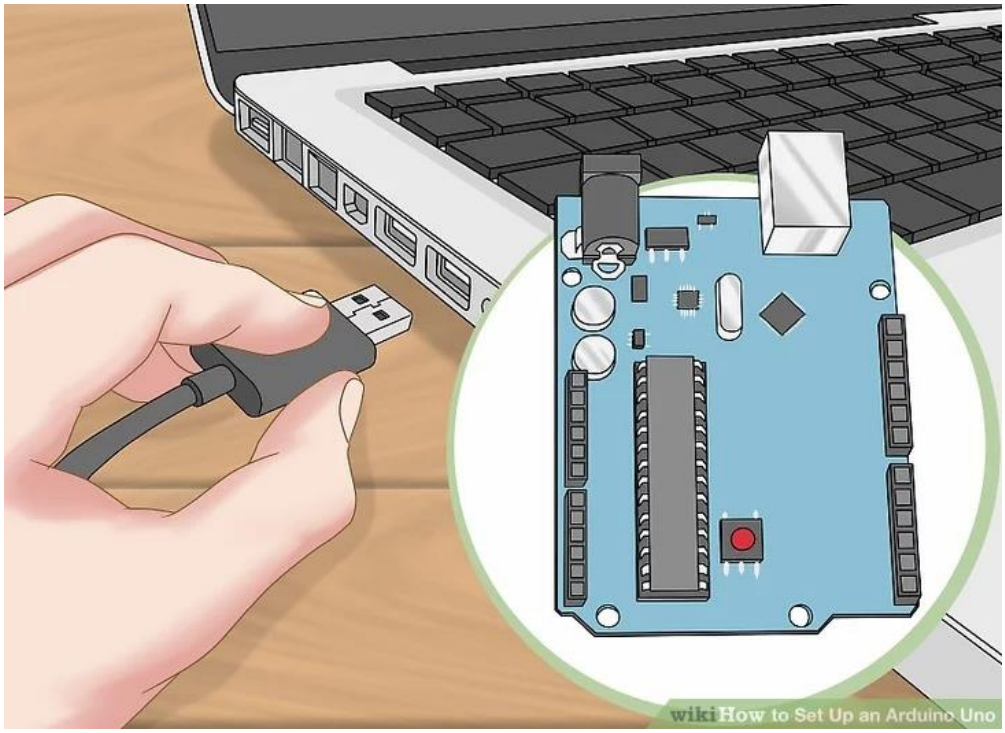
Table: Cost estimation of software

ARDUINO IDE INSTALLATION

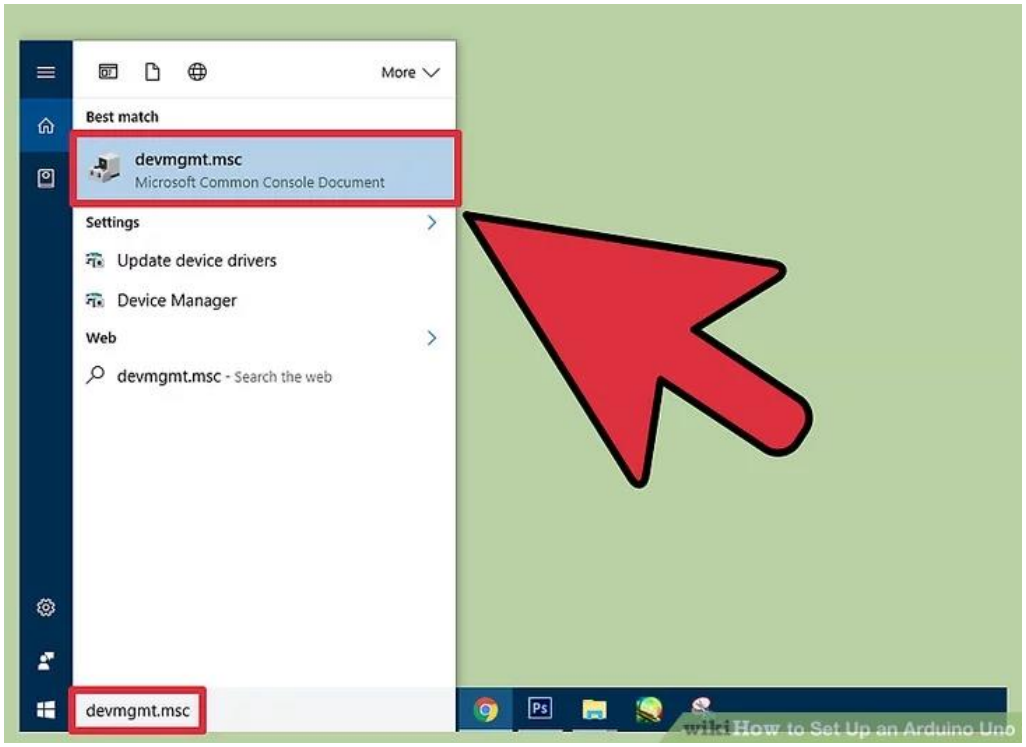
1.Download the latest version of the Arduino package and unzipping it. It may be around an 80MB download, fully expanded to 230MB.



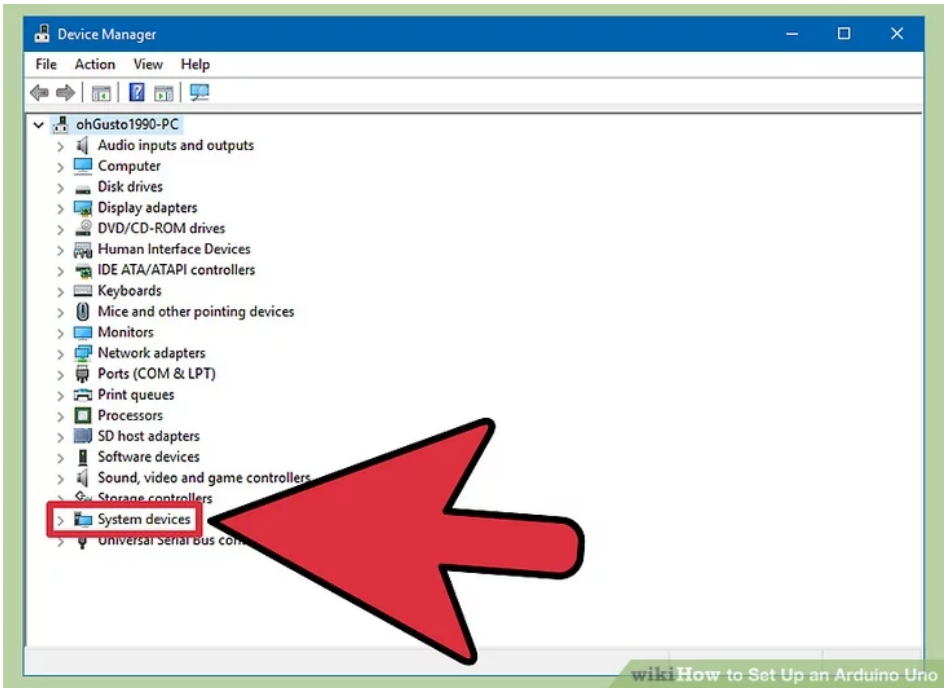
2.Plug the Arduino into your PC with the included USB cable. Close any pop-up windows and decline any search for drivers through Windows' tool.



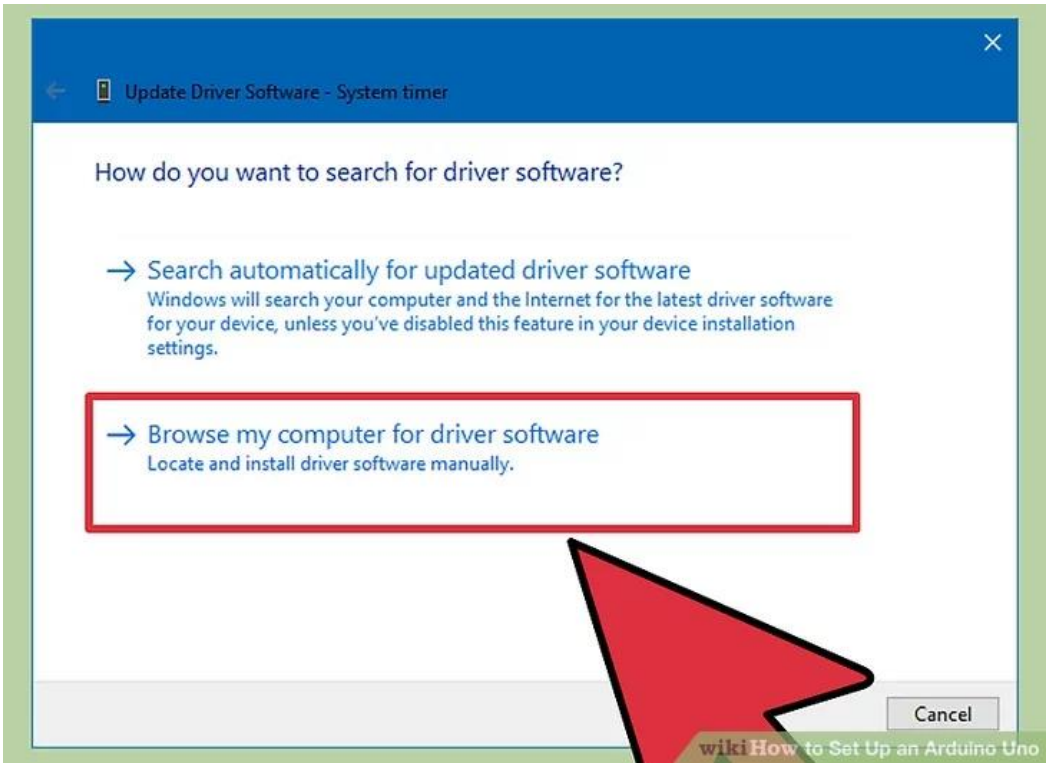
3.Click on “Start,” and then type devmgmt.msc. Press ↵ Enter and the Device Manager panel should open.



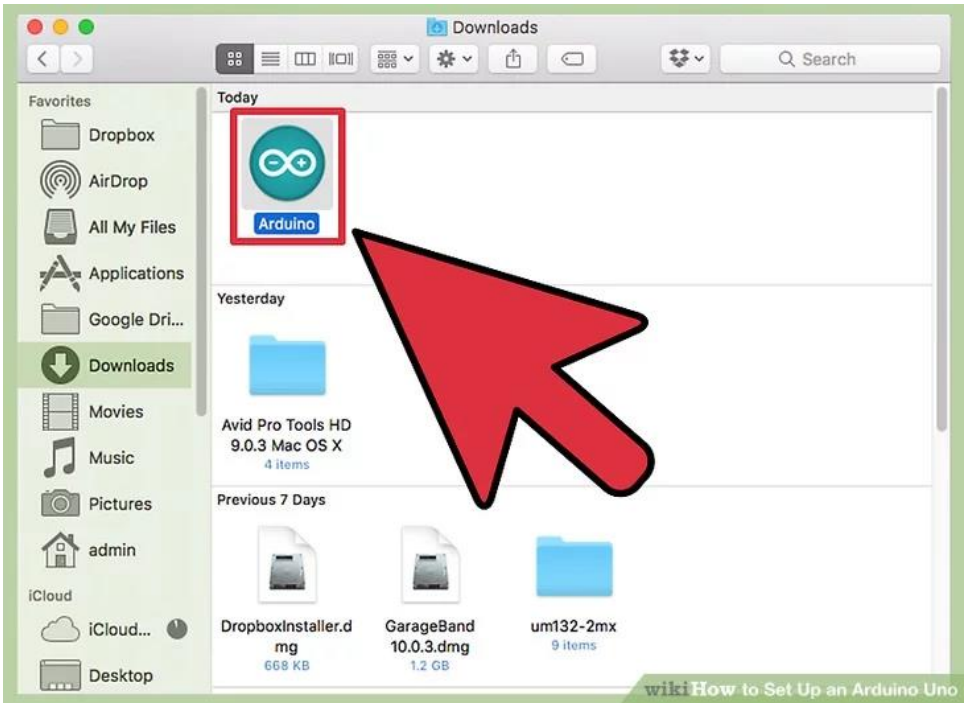
4.Find the ‘Other devices’ section. You should see an Arduino Uno on the list, with one of those yellow "problem" indications. Double-click on that, select ‘Update Drivers’ and then select ‘Browse my computer for driver software.’



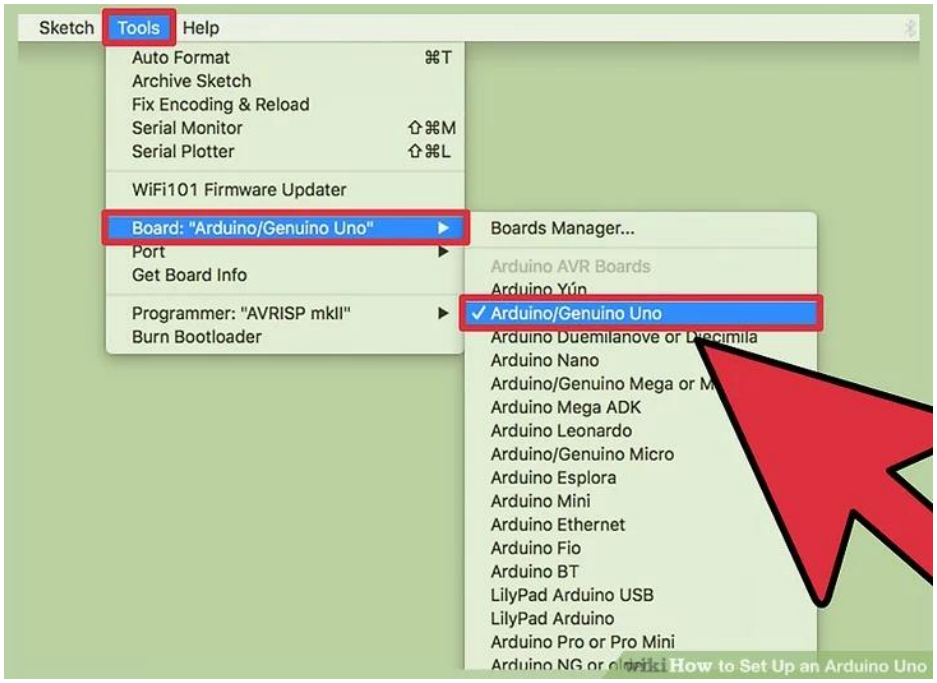
5.Select ‘Let me choose a Driver,’ click on ‘All Devices’, ‘Have Disk,’ and then go to the ‘Downloads’ folder. In the drivers folder, you should see an Arduino Uno file. Select that. The driver installation process will deem it a "communications port" and report a "mistake unsigned driver" when installing. Ignore that, proceeding with the installation anyway.



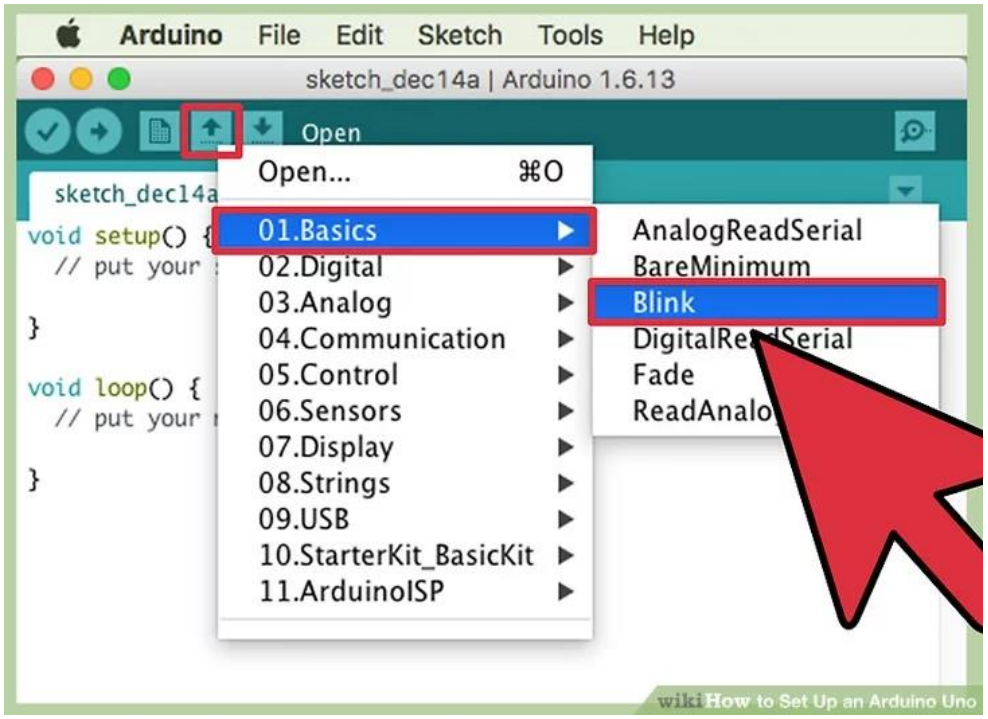
6.Launch the Arduino application from either the ‘Download’ folder or the ‘Applications’ folder.



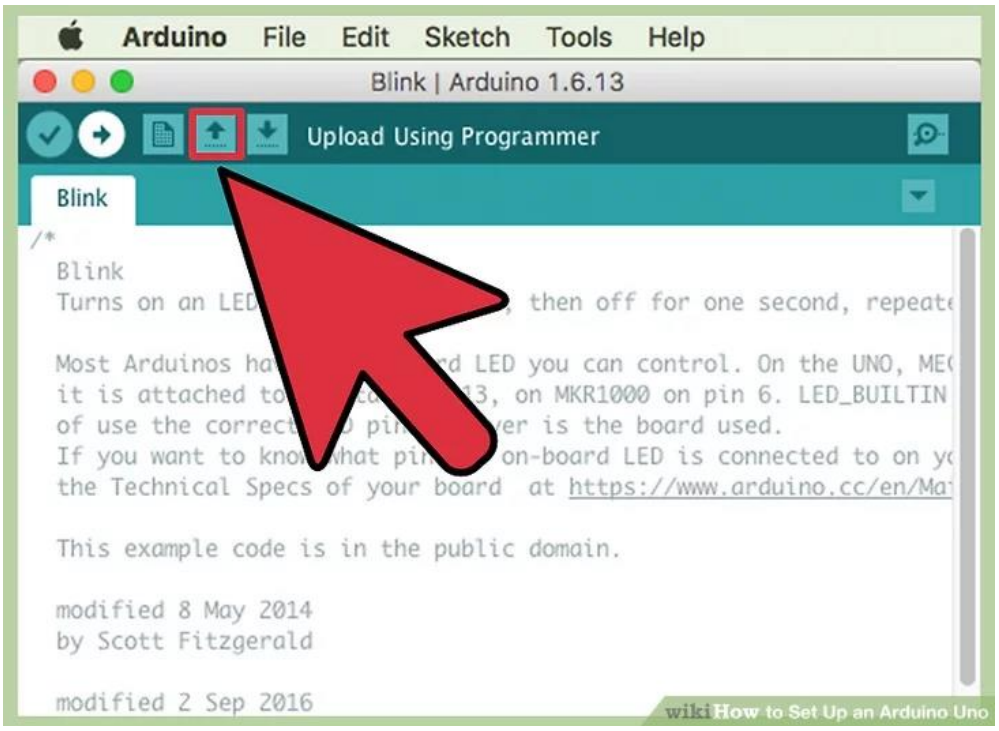
7.Set up the port and board. On either system, you may have a tools menu. Click that and ensure a matching board is chosen. The Uno is the default and may already be highlighted in your installation.



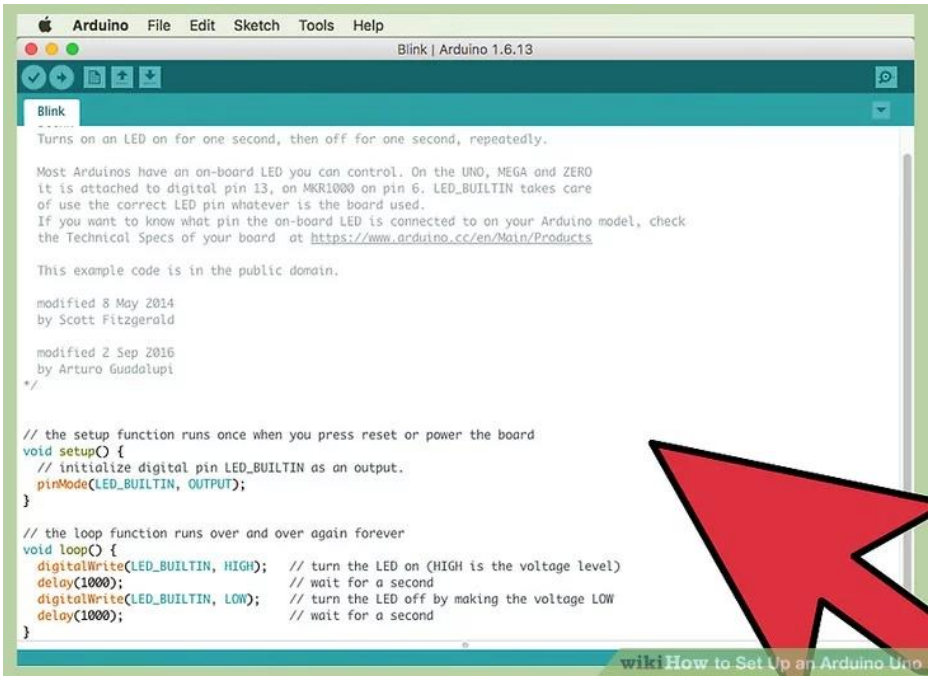
8.Open Arduino IDE,Click ‘Open,’ and from the dialog select the example application: Basics -> Blink.



9.Test the setup. To ensure that the board is working and is connected correctly, go ahead and click ‘Upload.’

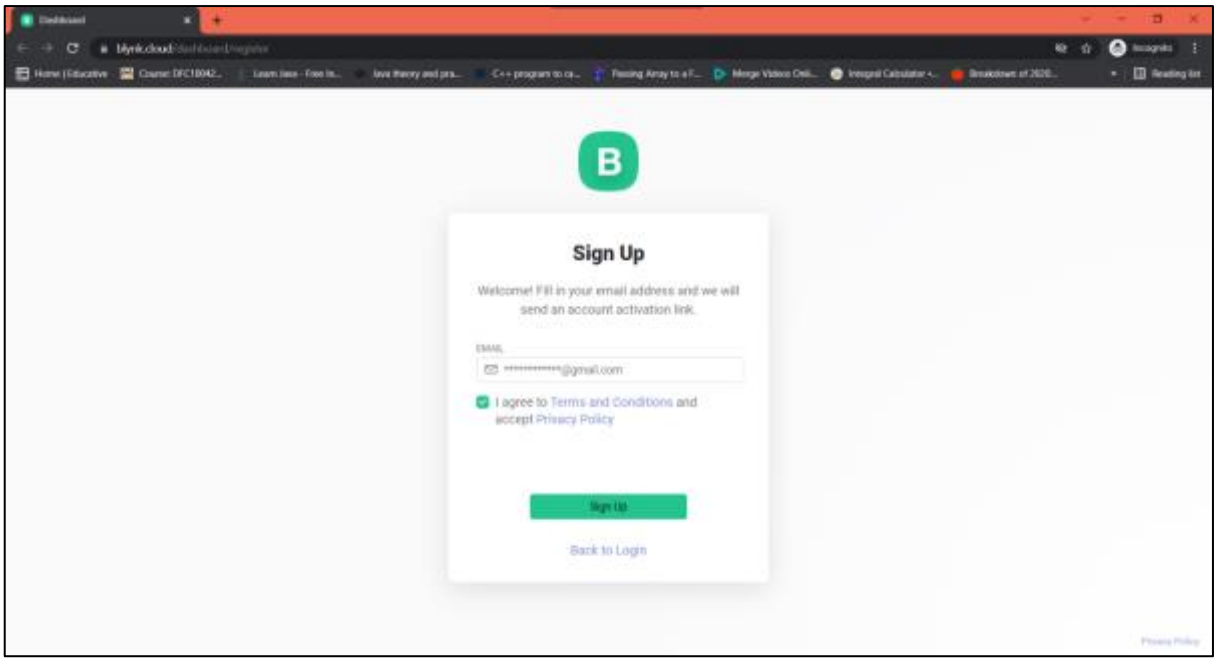


10.Start experimenting. That is as far as this tutorial will take you. You’ve now set up your Arduino and figured out how to upload apps to it, confirming that it works properly.



BLYNK REGISTRATION

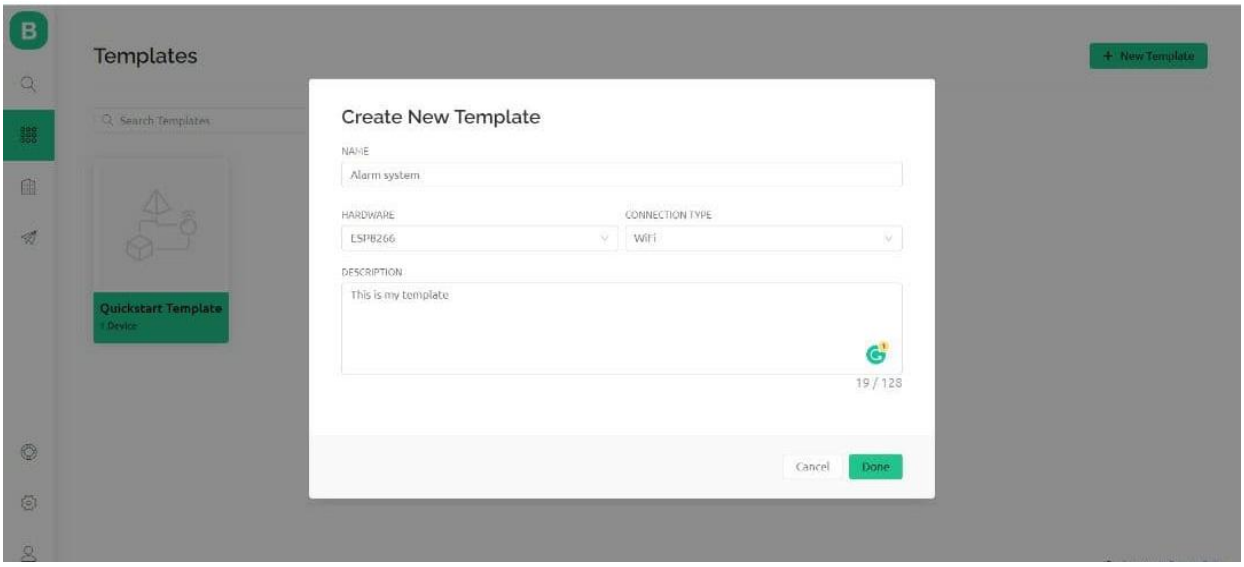
1. Register a free account on Blynk.



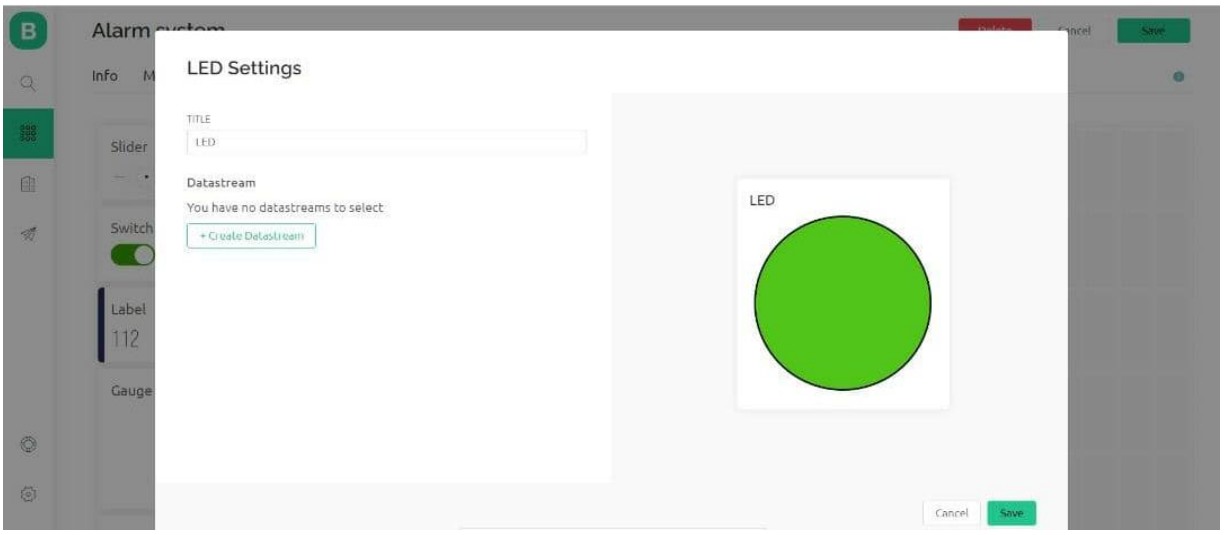
2. Log in and creating a device profile on Blynk Console.

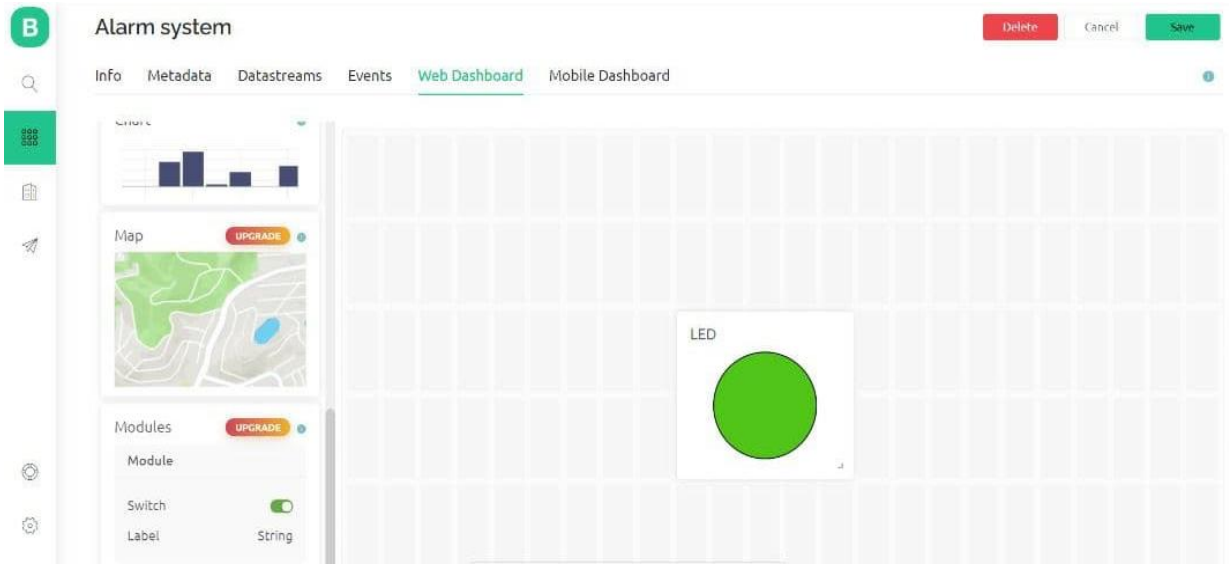


3. Then create the new template.

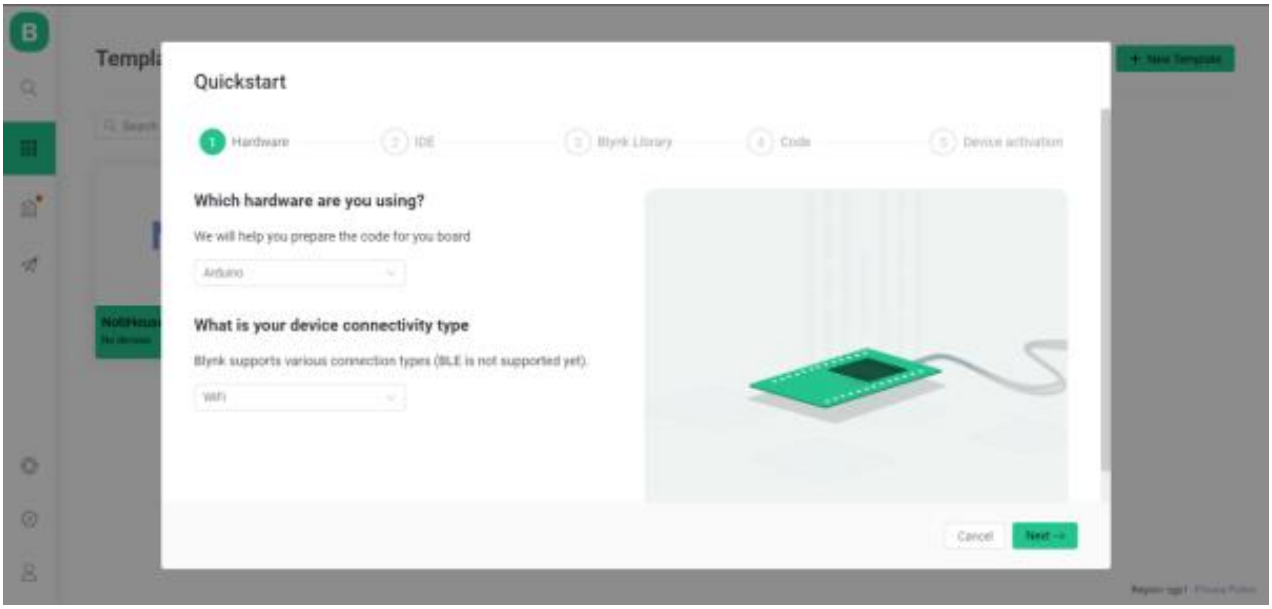


4. After that, design the GUI and drag and drop the widget

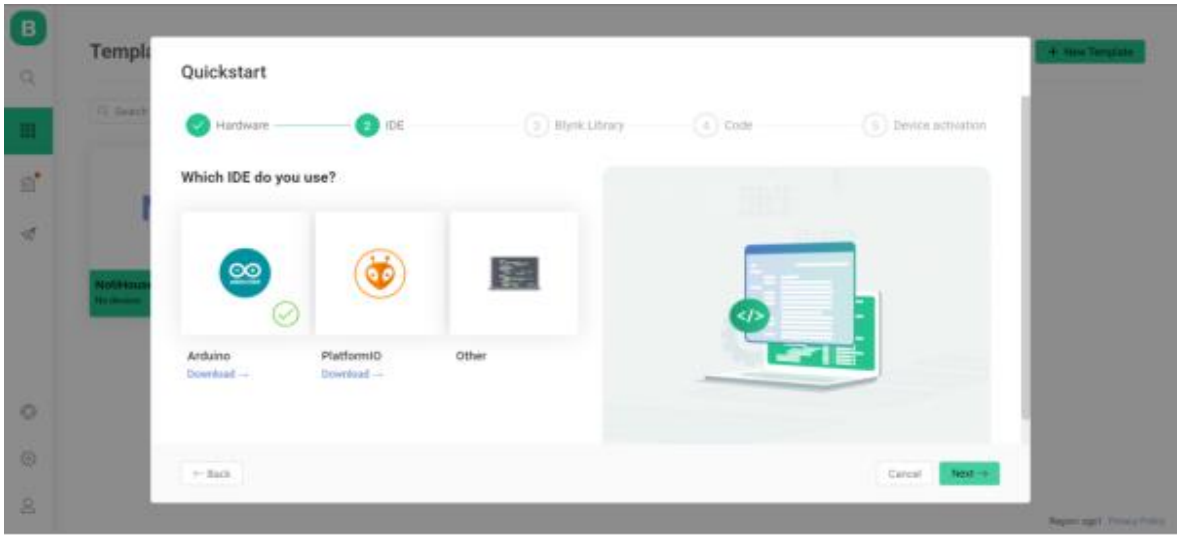




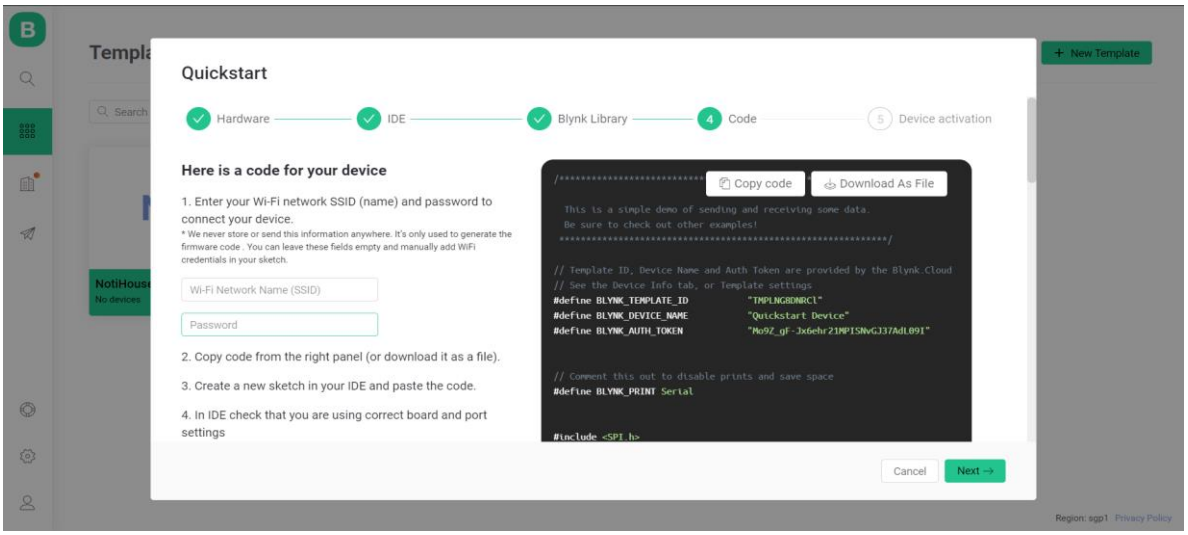
5. Specify device type and connectivity



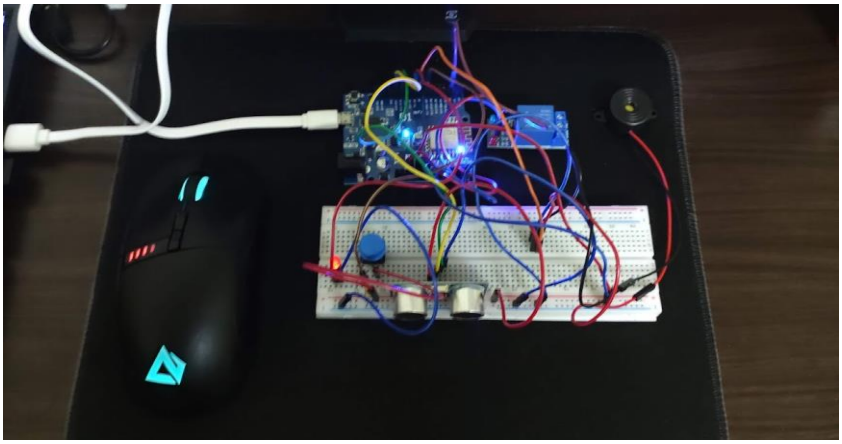
6. We choose preferred IDE



7. Copy Blynk Server connection code and paste into sketch code already written in Arduino IDE.

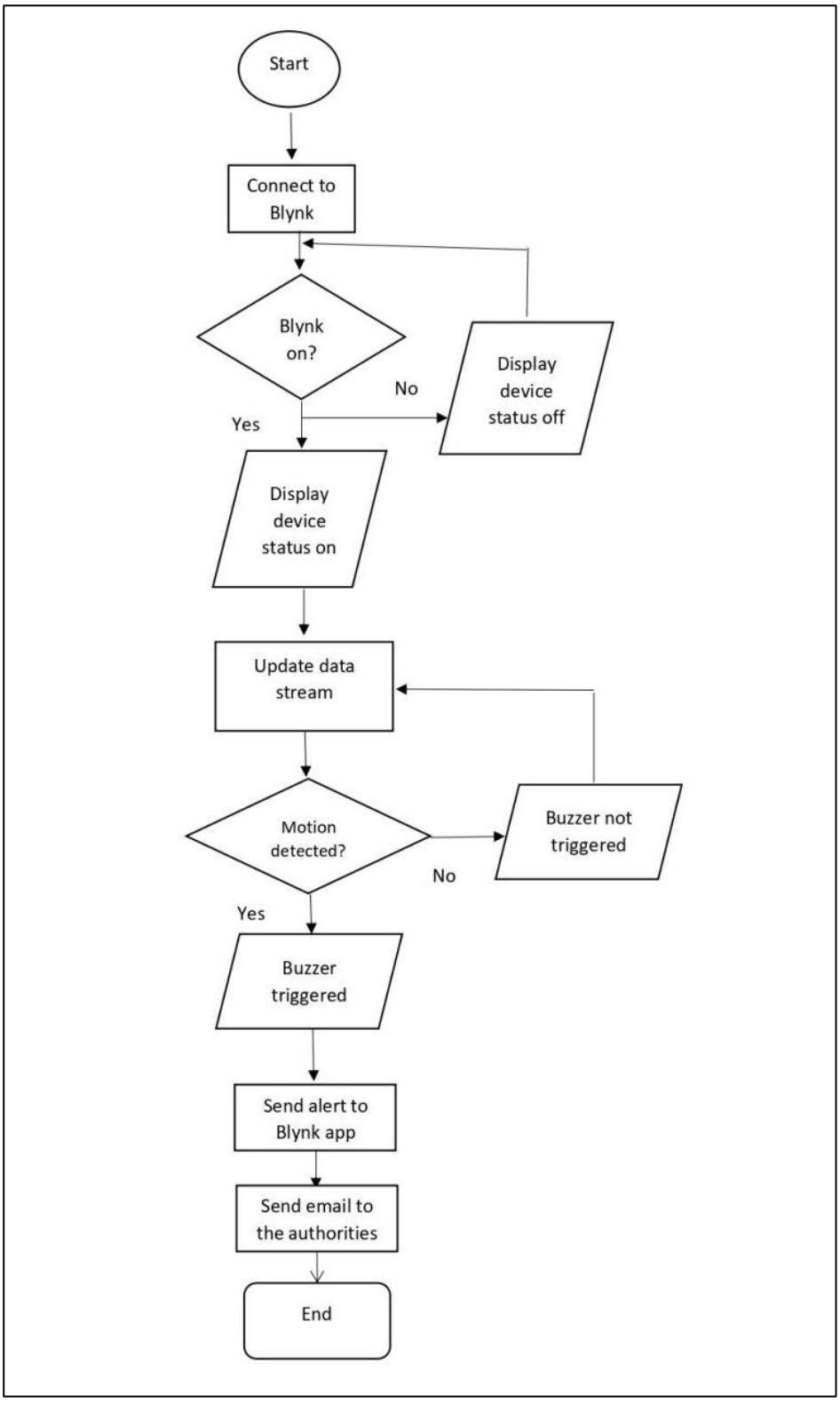


8. Lastly, connect device to PC to upload code.

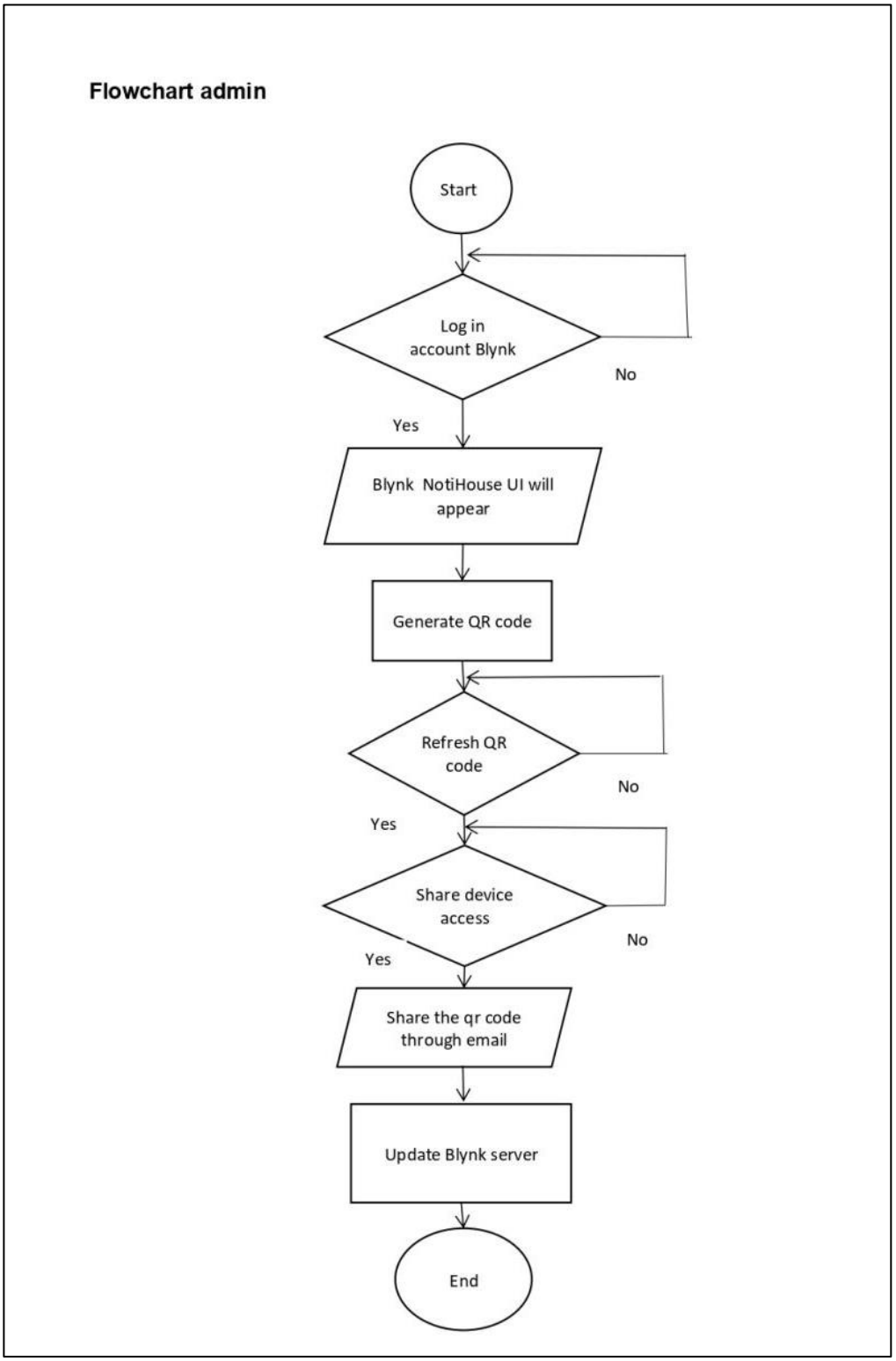


FLOWCHART

FLOWCHART FOR BLYNK APP



FLOWCHART FOR ADMIN



FLOWCHART FOR USER

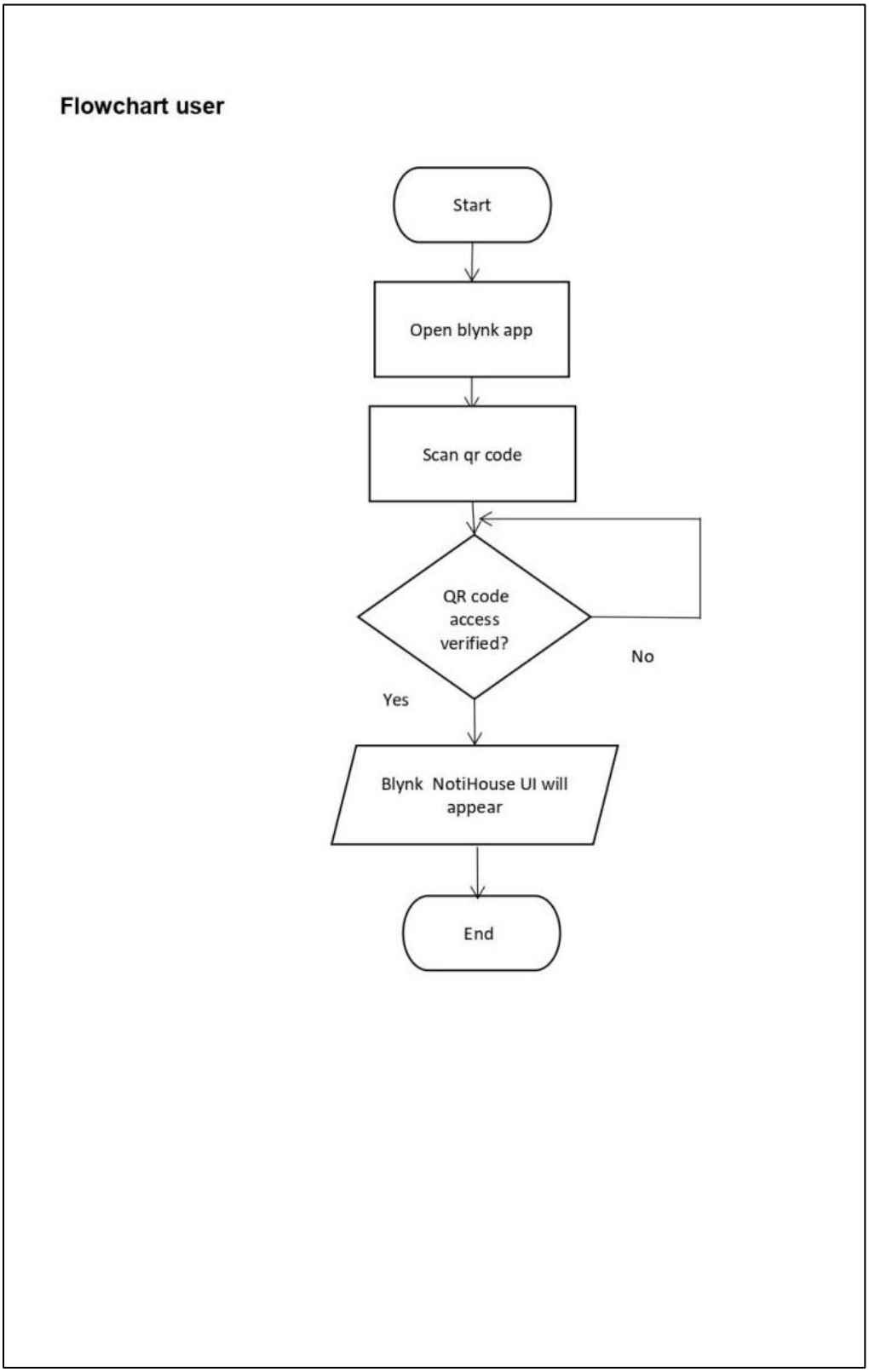


TABLE TESTING

UNIT TESTING

UNIT TESTING FOR NOTIHOUSE DEVICE

UNIT TESTING PLAN(UTP)						
No	Test Case Name	Test Procedure	Pre-Condition	Expected Result	Tester	Result(Pass /Fail)
1.	Establish internet connection	-turn on device from the Blynk app	Device must be connected to a power source	Blynk app indicate alarm system connected	Syafrul	Pass
2.	Ultrasonic sensor movement detection	Move an object / hand towards the sensor	Sketch code must already be uploaded from Arduino IDE to device	Change of measurement readings on serial monitor.	Syafrul	Pass
3.	Buzzer alarm beep	Move an object / hand towards the sensor	Sketch code must already be uploaded from Arduino IDE to device	Beeping sound produced by buzzer.	Adli	Pass
4.	Device on/off indication	Connect device to a power source.	Sketch code must already be uploaded from Arduino IDE to device	LED lights up and blinking if movements are detected	Adli	Pass

UNIT TESTING FOR BLYNK APP (ADMIN)

UNIT TESTING PLAN(UTP)						
No	Test Case Name	Test Procedure	Pre-Condition	Expected Result	Tester	Result(Pass/ Fail)
1.	Admin registration	Enter new username and password	Admin must fill up new username and password	Sign in page will be displayed	Adli	Pass
2.	Admin Login	Enter username and password	Admin must enter valid username and password	Designed UI in Blynk will be displayed	Adli	Pass
3.	Display push notification	Move an object or hand near the range of device	Motion detected by NotiHouse device	Notification will appear on user phone	Adli	Pass
4.	Send email notification	Move an object or hand near the range of device	Motion detected by NotiHouse device	Blynk will send an alert email to the authorities through email	Syafrul	Pass
5.	Display led indicator	Move an object or hand near the range of device	Motion detected and LED on NotiHouse device blinking.	Led indicator on blynk template device will blinking	Syafrul	Pass
6.	Remove user access	Regenerate access QR code	Admin must be logged in to Blynk account	User no longer have access to the app	Azni	Pass
7.	Shared user access	Share access QR code	Admin must be logged in to Blynk account	User now have access to the app	Syafrul	Pass

INTEGRATION TESTING

INTEGRATION TESTING FOR BLYNK APP (USER)

UNIT TESTING PLAN(UTP)						
No	Test Case Name	Test Procedure	Pre-Condition	Expected Result	Tester	Result(Pass/ Fail)
1.	Display push notification	Move an object or hand near the range of device	Motion detected by NotiHouse device	Notification will appear on user phone	Azni	Pass
2.	Send email notification	Move an object or hand near the range of device	Motion detected by NotiHouse device	Blynk will send an alert email to the authorities through email	Azni	Pass
3.	Display led indicator	Move an object or hand near the range of device	Motion detected and LED on NotiHouse device blinking.	Led indicator on blynk template device will blinking	Syafrul	Pass

INTEGRATION TESTING FOR LOGIN (USER)

INTEGRATION TESTING PLAN(ITP)						
No	Test Case Name	Test Procedure	Pre-Condition	Expected Result	Tester	Result(Pass/ Fail)
1.	User create the account	1. Open the Blynk apps and create the new account	1. Create name and password	Successfully to create the account	Adli	Pass
2.	Log in your account on Blynk Legacy to create new project	1. User can enter the Blynk with their account	Enter the password	User able to log in the app	Adli	Pass

APPENDIX 1 : User Manual

The user manual contains all essential information for the user to make full use of the NotiHouse Alarm Syatem. This user manual includes description of the system functions and step by step procedures for system access and use.

User Manual : Physical

1. Place the device somewhere strategic to detect as much motion as possible in a room. Recommended position (Above the door).

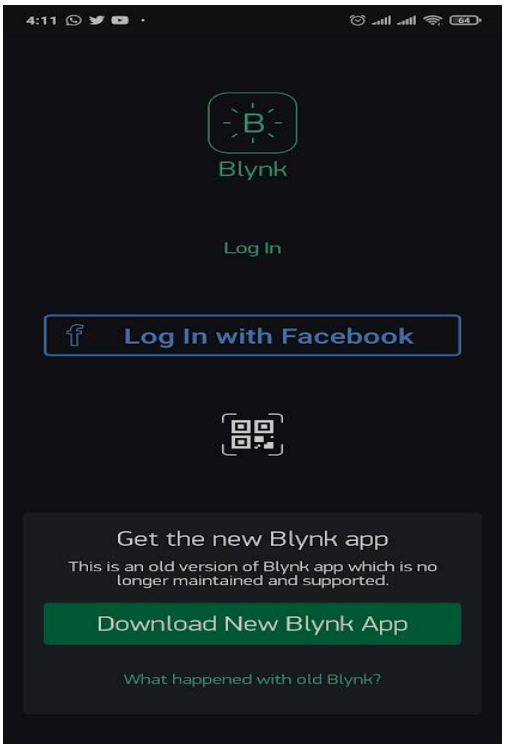


2. Plug-in the device to a power source.

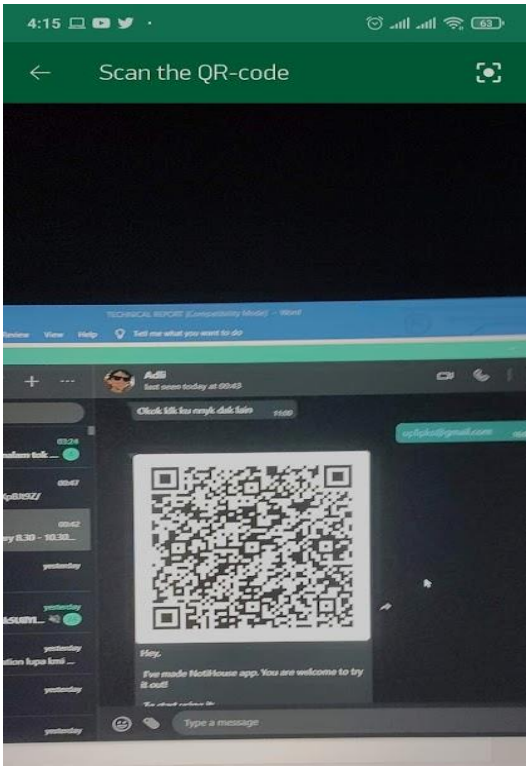


User Manual : Application(User)

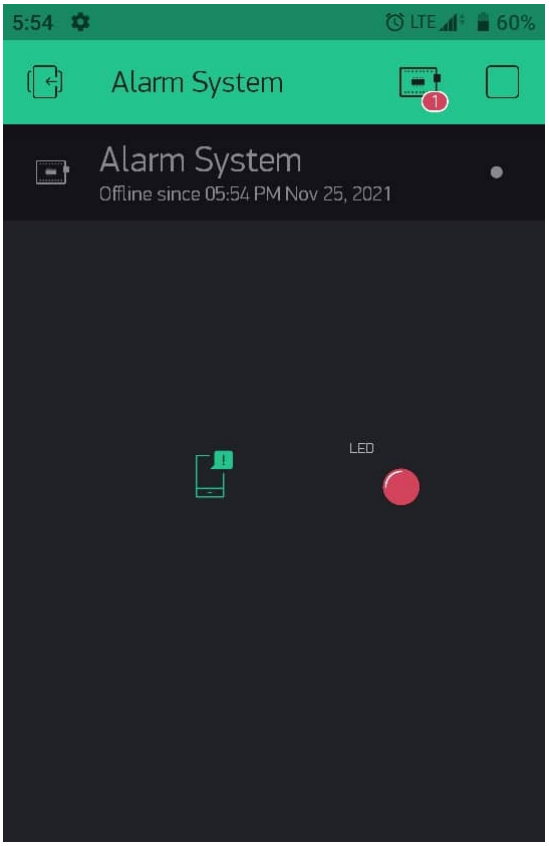
1. Turn on the device and start logging into the Blynk App.



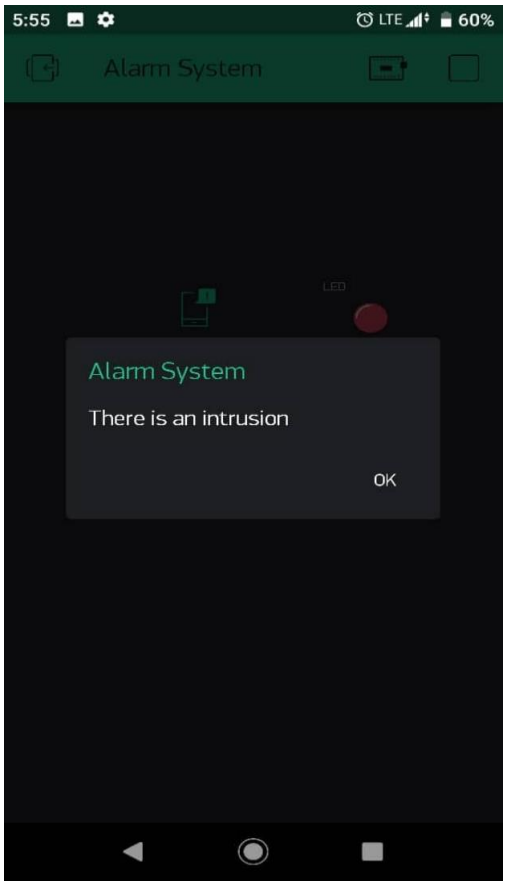
2. Click on the Scan Button in the center to Login into your account using the QR Code provided by the admin (us).



3. Once logged in, you are now able to monitor the device status.



4. You can now also receive notifications for any intrusion.

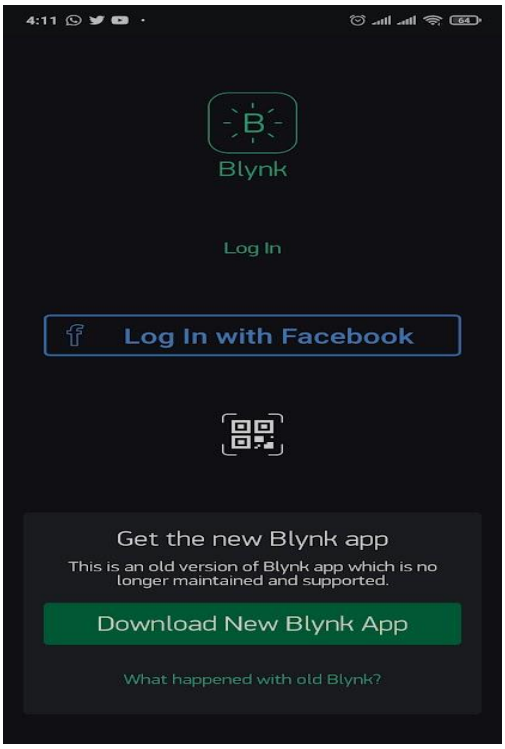


5. You will also be notified of intrusion through email. If somehow your phone is silent or blocked notifications.

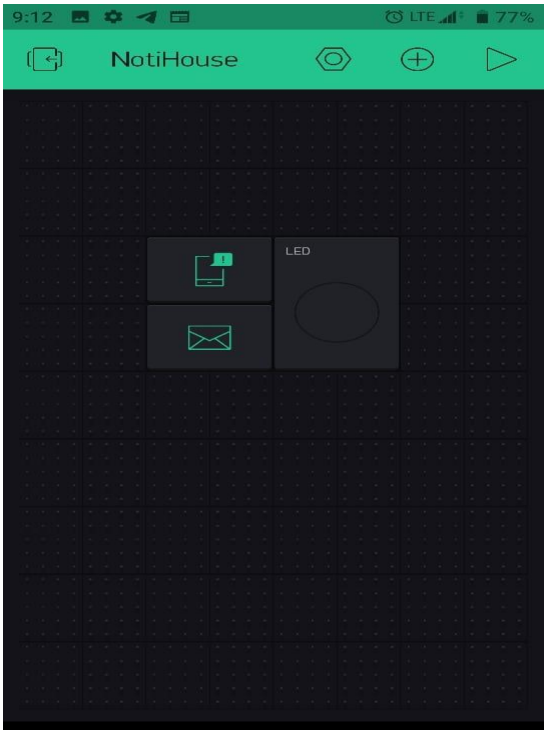


User Manual : Application(Owner/Developer)

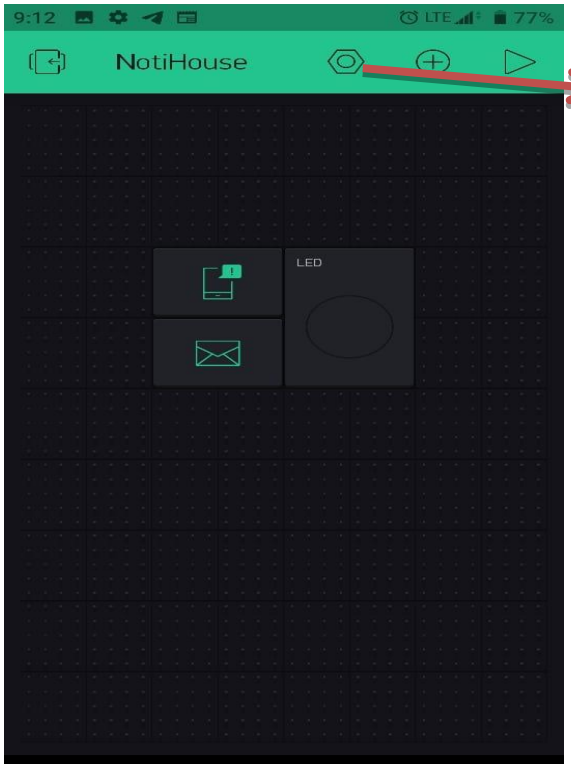
1. Turn on the device and start logging into the Blynk App.



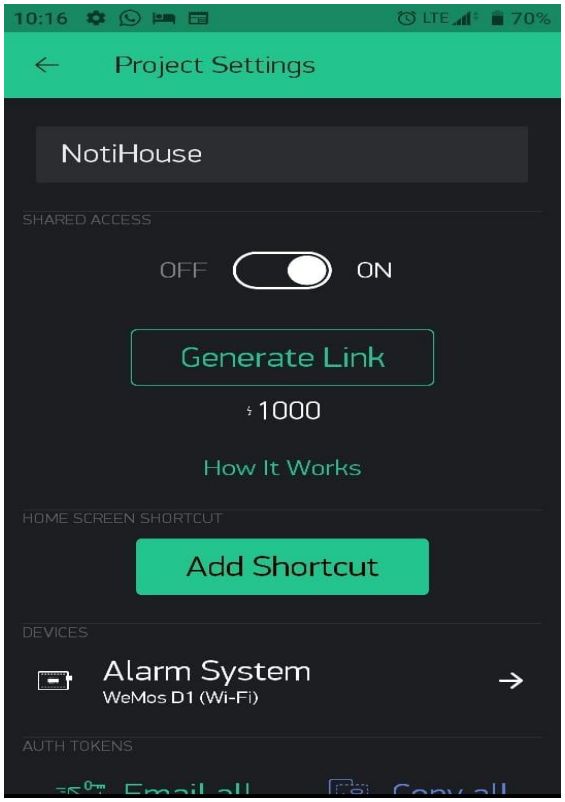
2. Blynk NotiHouse ui will appeared on the device



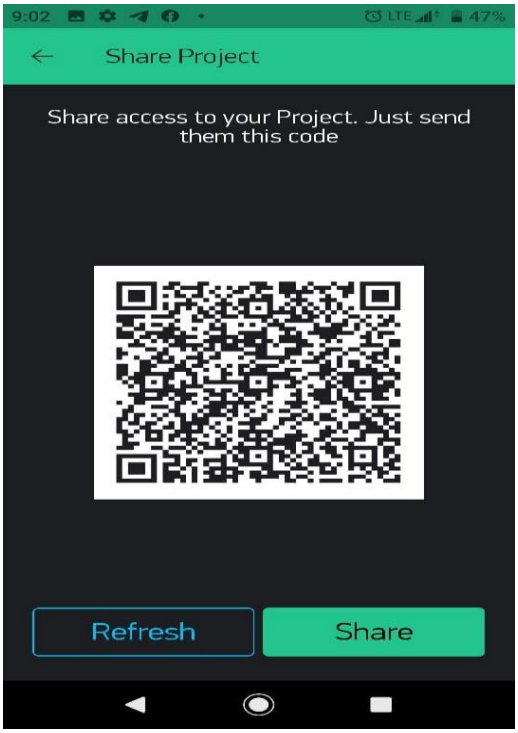
3. The owner can configure the project setting



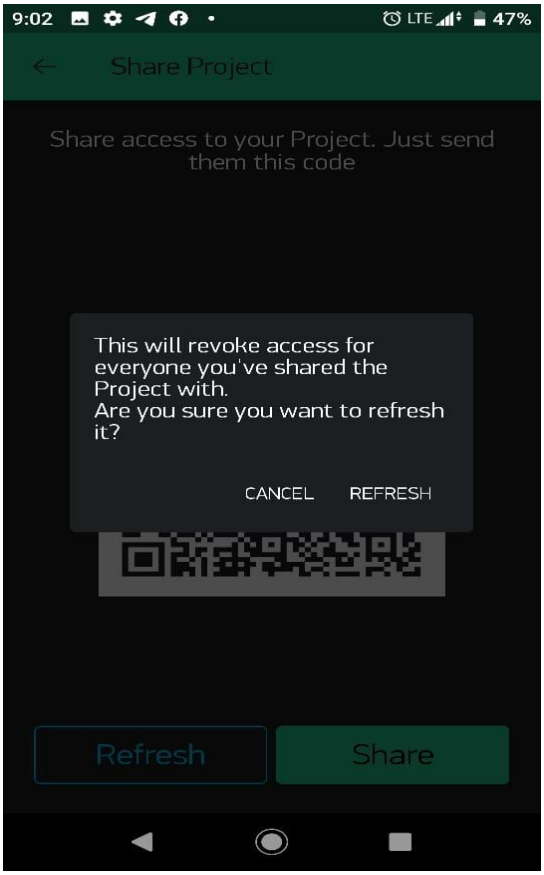
Click the button to configure the project setting



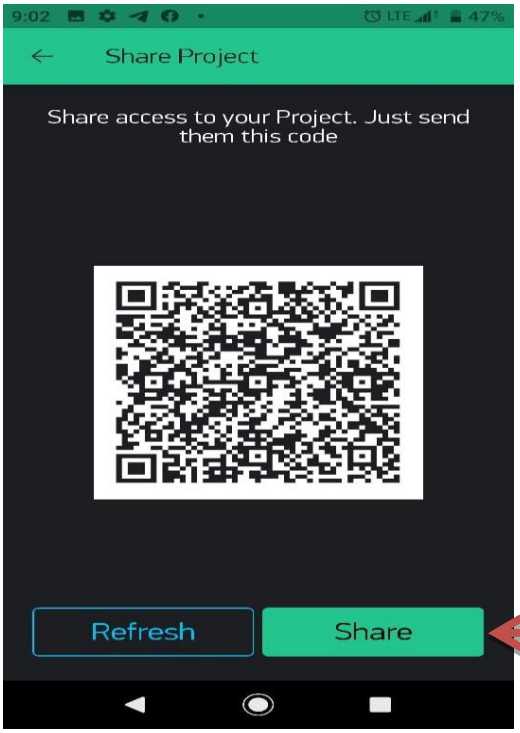
4. The owner can refresh or share the QR code to the user.



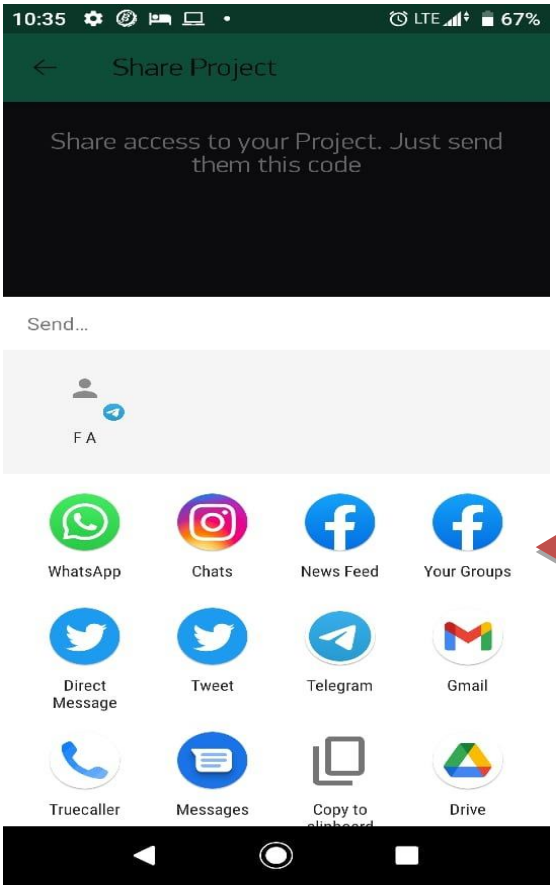
5. If the owner want to revoke the previous user access,then the owner will click refresh button.



6. The owner can share the new Qr code access to the user by clicking share button

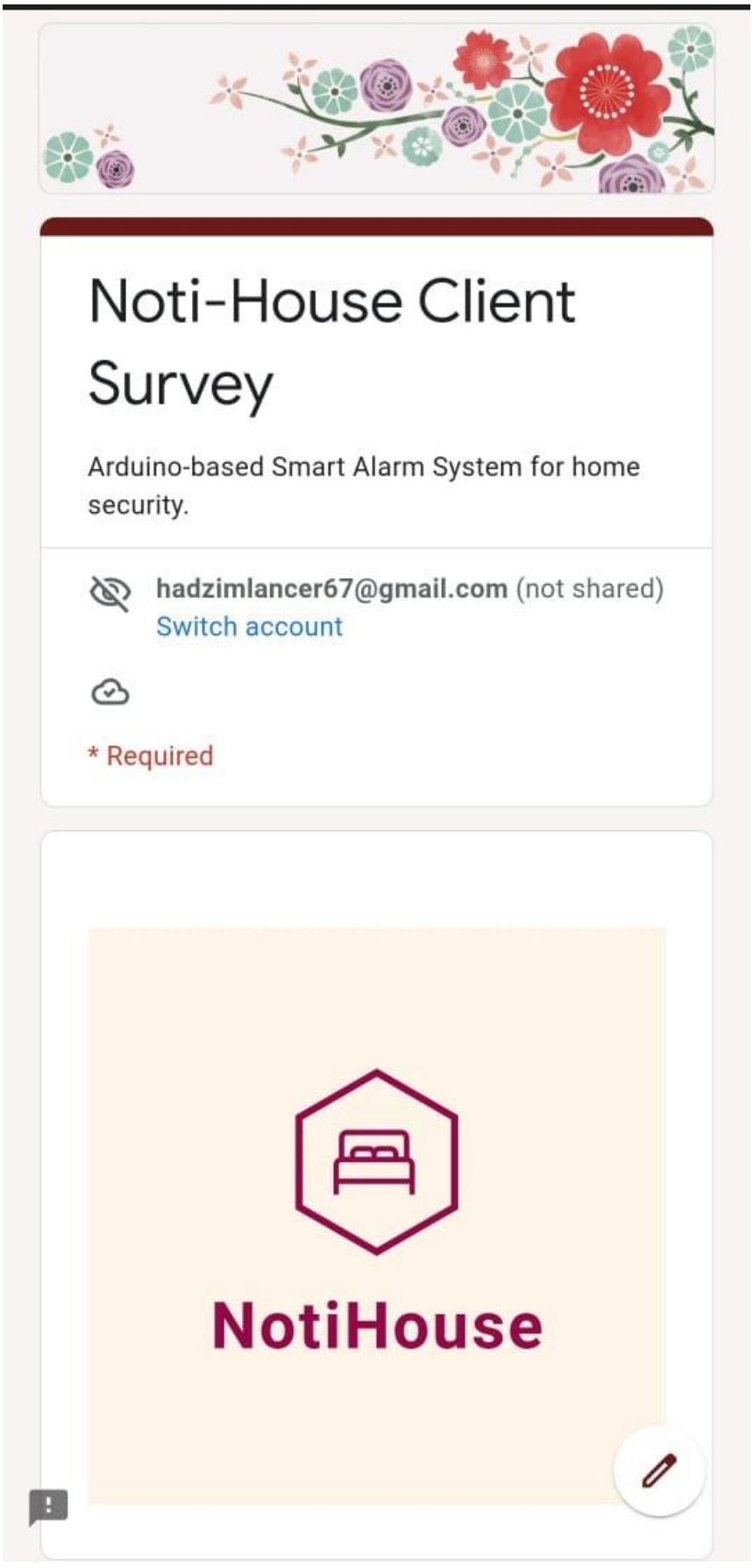


Click this button to share the new QR code



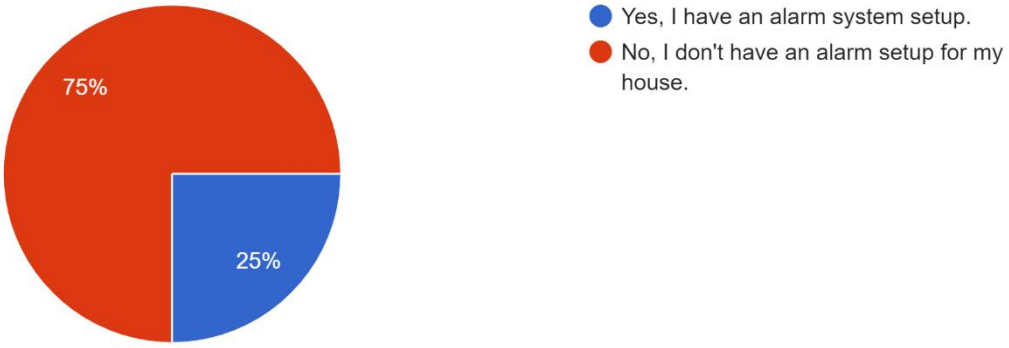
The owner can choose whichever platform that he want to share the Qr code to the user

APPENDIX 2 : Interview Question For Requirement Elicitation



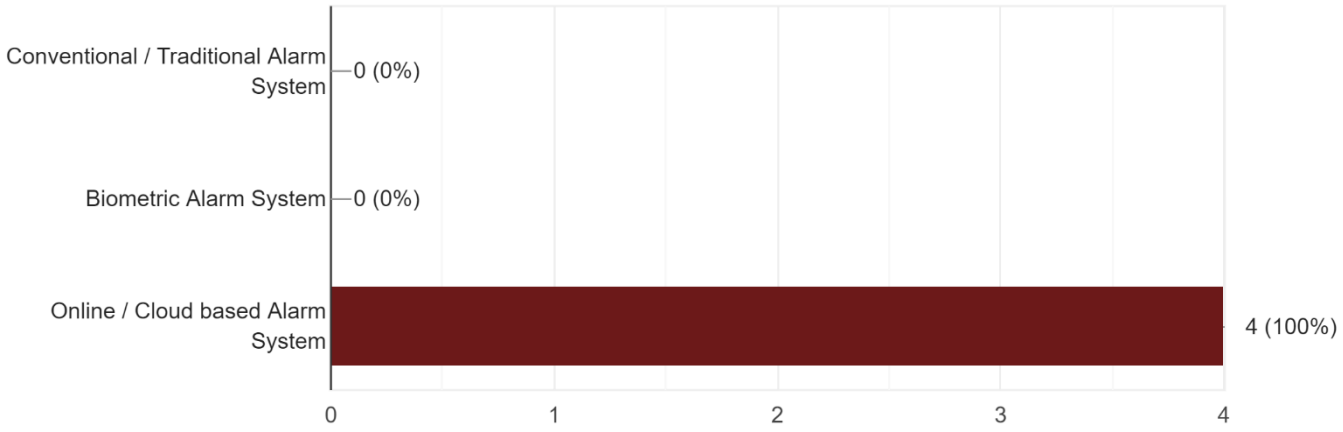
Do you have an alarm system setup for your house ?

4 responses



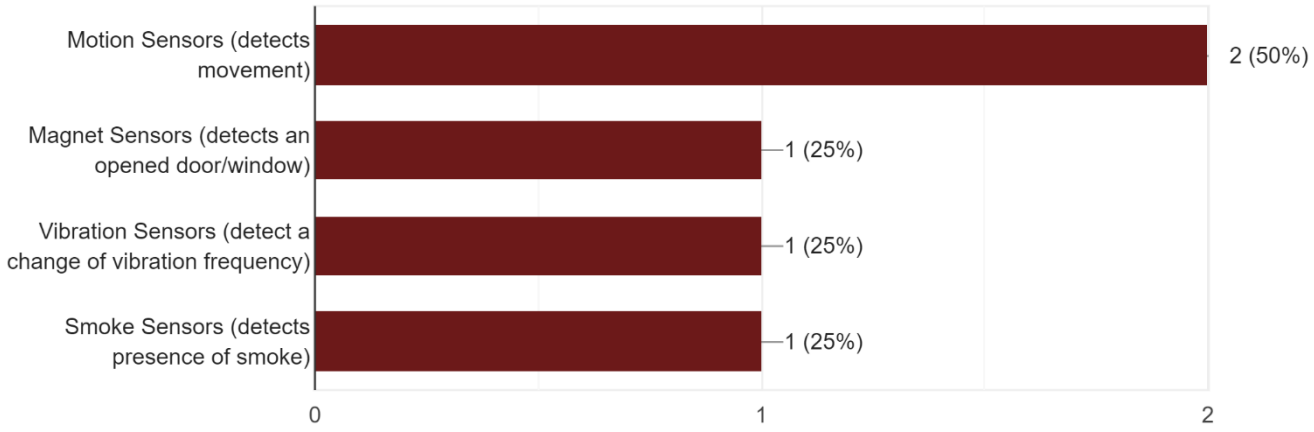
What type of alarm system do you have or would like to have ?

4 responses



What type of sensors do you think is important for an alarm system ?

4 responses



What is your ideal price for an alarm system ?

4 responses

100

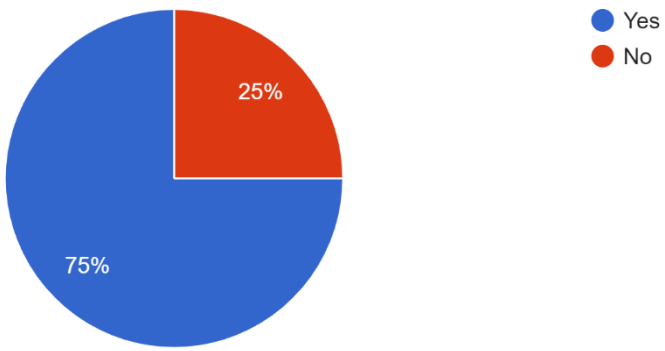
250

80-100

399

Would you like to know if there is an intrusion in your house immediately ?

4 responses



Why should house owners always be notified of an intrusion regardless if they are/not at home ?

4 responses

beacuse we do care about our house

To make sure their house is safe

to stay vigilant incase of who broke into the house and steal any of the belongings

for security/alert

Your name

4 responses

mohamad amirul

Nur Neelam

Anis Ayuni

Dayangku Intan

Contact info

E-mail

4 responses

aniqzzz1108@gmail.com

nneelam0808@gmail.com

anisayunihossin@gmail.com

dayangkuintan05@gmail.com

Preferred contact method

4 responses

